



# **ANOMALY DETECTION FOR FRAUD RISK IN BANKING DATA**

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# Table of Content

- 01 Context
- 02 EDA
- 03 Data Preprocessing
- 04 Modeling
- 05 Conclusion
- 06 Appendix



# Context

## Fraud Risk Detection

“All banks strive to ensure secure and trustworthy financial transactions.

Detecting and preventing fraudulent activity is essential to protect customers, maintain institutional integrity, and minimize financial losses.”



# Context



## Problem

Fraudulent transactions cause major financial losses and reduce customer trust. Detecting such rare anomalies is challenging.



## Objective

Detect suspicious transactions with data-driven methods to enhance fraud prevention and risk control.



## Methods

Dataset: [Bank Transaction Dataset for Fraud Detection](#)  
Tool: Python.

# Context

## TransactionID

Unique identifier for each transaction.

## AccountID

Unique customer account ID.

## TransactionAmount

Amount of money involved in the transaction.

## TransactionDate

Timestamp of the transaction.

## TransactionType

Type of transaction (`Credit`, `Debit`, etc.)

## Location

City where the transaction occurred.

## DeviceID

Unique identifier of the device used.

## IP Address

IP address from which the transaction originated.

## MerchantID

Unique ID for the merchant involved in the transaction.

## Channel

Channel used (`Online`, `ATM`, etc.)

## CustomerAge

Age of the customer.

## CustomerOccupation

Occupation of the customer (`Doctor`, `Student`, etc.)

## TransactionDuration

Duration (in seconds) of the transaction session.

## LoginAttempts

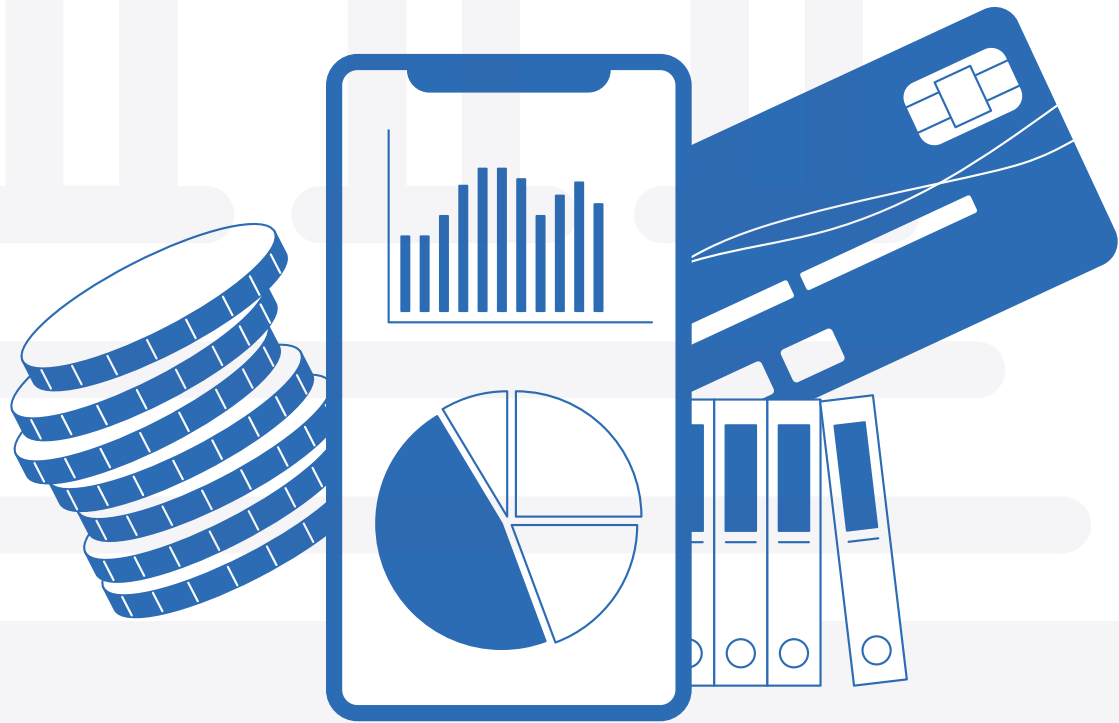
Number of login attempts before the transaction.

## AccountBalance

Account balance after the transaction.

## PreviousTransactionDate

Timestamp of the customer's previous transaction.



# EDA – Data Preview and Infor

## Data information

Data columns (total 16 columns):

| #  | Column                  | Non-Null Count | Dtype   |
|----|-------------------------|----------------|---------|
| 0  | TransactionID           | 2512 non-null  | object  |
| 1  | AccountID               | 2512 non-null  | object  |
| 2  | TransactionAmount       | 2512 non-null  | float64 |
| 3  | TransactionDate         | 2512 non-null  | object  |
| 4  | TransactionType         | 2512 non-null  | object  |
| 5  | Location                | 2512 non-null  | object  |
| 6  | DeviceID                | 2512 non-null  | object  |
| 7  | IP Address              | 2512 non-null  | object  |
| 8  | MerchantID              | 2512 non-null  | object  |
| 9  | Channel                 | 2512 non-null  | object  |
| 10 | CustomerAge             | 2512 non-null  | int64   |
| 11 | CustomerOccupation      | 2512 non-null  | object  |
| 12 | TransactionDuration     | 2512 non-null  | int64   |
| 13 | LoginAttempts           | 2512 non-null  | int64   |
| 14 | AccountBalance          | 2512 non-null  | float64 |
| 15 | PreviousTransactionDate | 2512 non-null  | object  |

dtypes: float64(2), int64(3), object(11)  
memory usage: 314.1+ KB

## Checking duplicate

=====

🔍 Missing Value Summary for: Dataset

=====

✅ No missing values detected in 2,512 rows.

## Checking outlier

=====

🔍 Data Checking outlier

=====

|   | Feature           | Outlier Count |
|---|-------------------|---------------|
| 0 | TransactionAmount | 113           |



## Initial Data Exploration

### Dataset Overview

- 2512 records × 16 columns
- No target column.

### Feature Summary

- Numerical Features:
  - **TransactionAmount, CustomerAge, TransactionDuration, AccountBalance.**
  - **, age, tenure, balance, estimated\_salary.**
- Categorical Features:
  - **TransactionID, AccountID, DeviceID, MerchantID, TransactionDate, PreviousTransactionDate, TransactionType, Channel, Location, IP Address, CustomerOccupation, LoginAttempts.**

### ✅ Data Quality

- No missing values detected.
- No duplicate value detected.
- Outlier occurs at features **TransactionAmount.**

# EDA – Data Describe

Data describe numerical features:

|                     | count       | mean        | std         | min        | 25%         | 50%         | 75%         | max          |
|---------------------|-------------|-------------|-------------|------------|-------------|-------------|-------------|--------------|
| TransactionAmount   | 2512.000000 | 297.593778  | 291.946243  | 0.260000   | 81.885000   | 211.140000  | 414.527500  | 1919.110000  |
| CustomerAge         | 2512.000000 | 44.673965   | 17.792198   | 18.000000  | 27.000000   | 45.000000   | 59.000000   | 80.000000    |
| TransactionDuration | 2512.000000 | 119.643312  | 69.963757   | 10.000000  | 63.000000   | 112.500000  | 161.000000  | 300.000000   |
| AccountBalance      | 2512.000000 | 5114.302966 | 3900.942499 | 101.250000 | 1504.370000 | 4735.510000 | 7678.820000 | 14977.990000 |

- **TransactionAmount:** Avg  $\approx$  298 — range 0.26 – 1919, concentrated between 82 – 415.
- **CustomerAge:** Avg  $\approx$  45 years — range 18 – 80, most customers in 27 – 59.
- **TransactionDuration:** Avg  $\approx$  120 s — range 10 – 300, centered around 60 – 160 s.
- **AccountBalance:** Avg  $\approx$  3901 — range 101 – 14,978, majority between 1504 – 7679.

✔ **TransactionAmount** and **AccountBalance** show the highest variability, indicating potential anomalous or high-risk transaction behaviors worth deeper fraud risk analysis.

Data describe categorical features:

|                    | count | unique | top            | freq |
|--------------------|-------|--------|----------------|------|
| TransactionID      | 2512  | 2512   | TX002496       | 1    |
| AccountID          | 2512  | 495    | AC00460        | 12   |
| DeviceID           | 2512  | 681    | D000663        | 9    |
| MerchantID         | 2512  | 100    | M026           | 45   |
| TransactionType    | 2512  | 2      | Debit          | 1944 |
| Channel            | 2512  | 3      | Branch         | 868  |
| Location           | 2512  | 43     | Fort Worth     | 70   |
| IP Address         | 2512  | 592    | 200.136.146.93 | 13   |
| CustomerOccupation | 2512  | 4      | Student        | 657  |
| LoginAttempts      | 2512  | 5      | 1              | 2390 |

- **TransactionType:** Highly imbalanced — Debit transactions dominate (~77%), driving most activities.
- **Channel:** Three channels used; Branch is most common, followed by Online and ATM.
- **CustomerOccupation:** Students form the largest group, indicating a young, active customer base.
- **LoginAttempts:** Mostly single login (95%), suggesting normal access behavior.
- **Location:** 43 cities — geographically diverse, with Fort Worth most frequent.
- **MerchantID / AccountID:** Some merchants and accounts are much more active, potential focus points.
- **DeviceID & IP Address:** High uniqueness across records — valuable for identifying fraudulent or multi-account activity.

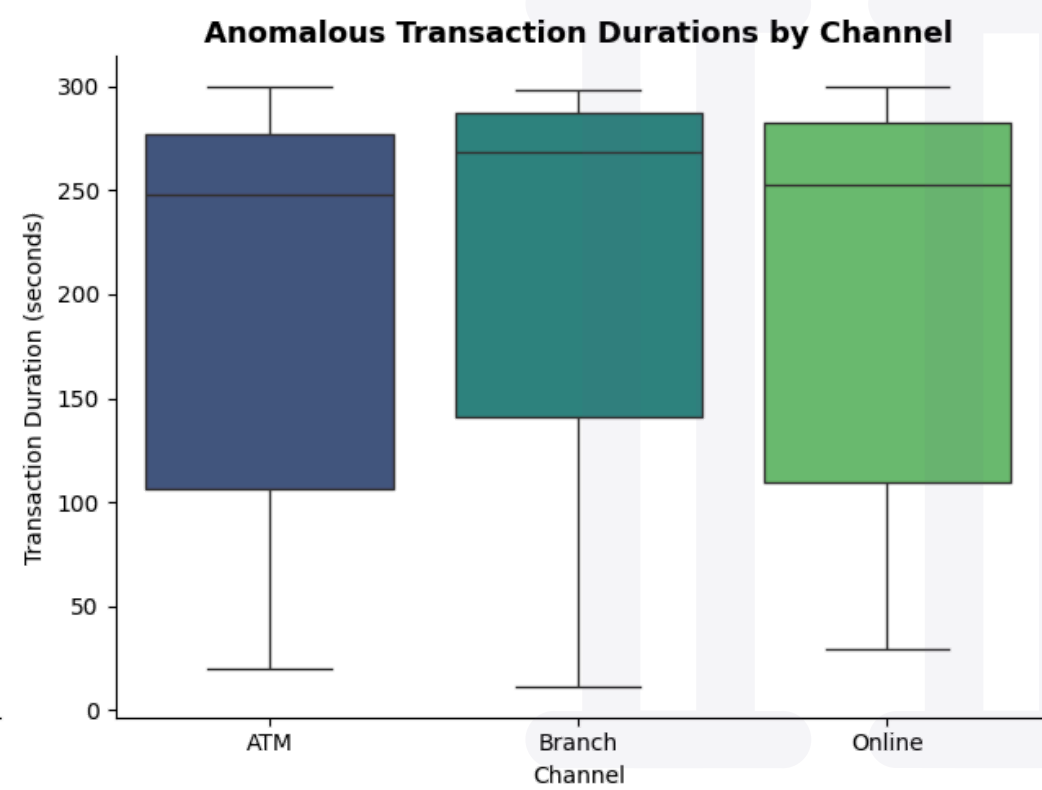
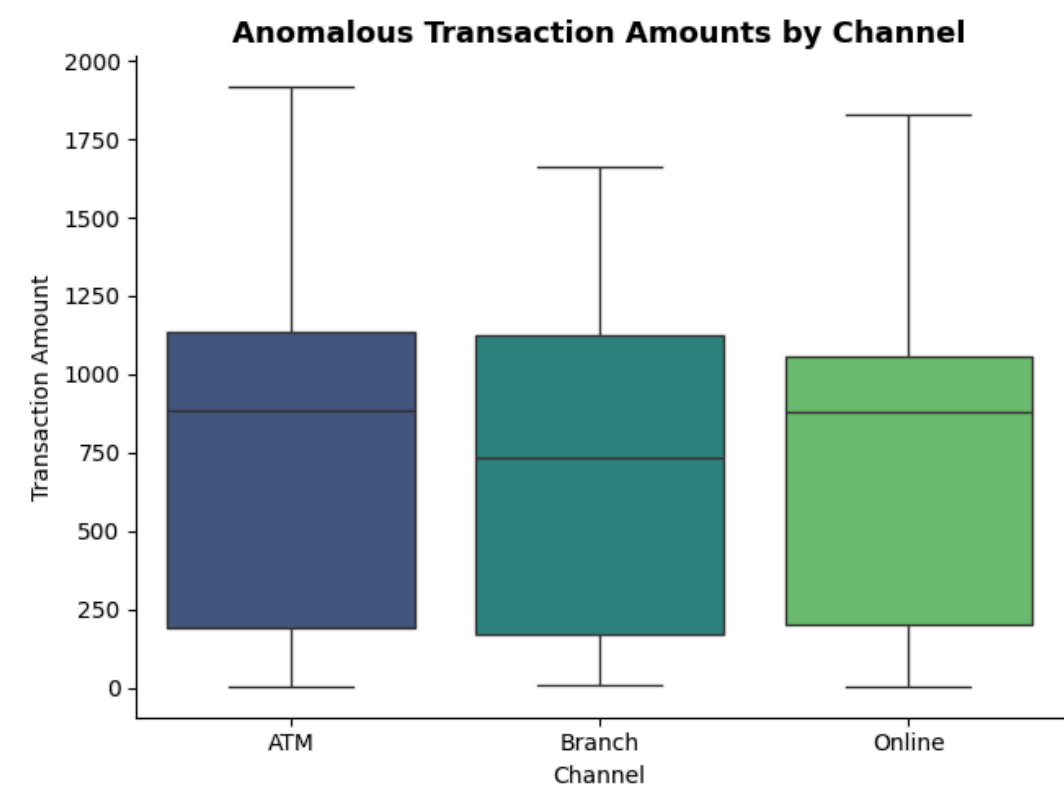


# EDA – Business Question

Which channels (ATM / Online / Branch) are associated with the largest anomalies in transaction amount or duration?

- Number of anomalous transactions: **250**
- Amount threshold (95th percentile): **878.18**
- Duration threshold (95th percentile): **265.00**

|   | Channel | AvgAmount  | MedianAmount | MaxAmount | AvgDuration | MedianDuration | MaxDuration | Count |
|---|---------|------------|--------------|-----------|-------------|----------------|-------------|-------|
| 0 | Online  | 688.963111 | 880.285      | 1831.02   | 193.622222  | 253.0          | 300         | 90    |
| 1 | ATM     | 738.663647 | 885.330      | 1919.11   | 191.741176  | 248.0          | 300         | 85    |
| 2 | Branch  | 681.003733 | 734.860      | 1664.33   | 215.746667  | 268.0          | 298         | 75    |



## Answer

👉 **Digital channels (ATM & Online)** are more associated with large or long-duration anomalies – implying greater operational risk and fraud exposure compared to in-branch transactions.

# EDA – Business Question

## Are there IPs or devices shared by multiple accounts?

🔑 Amount threshold (95th percentile): 878.18  
🕒 Duration threshold (95th percentile): 265.00  
Number IP shared by multiple accounts : 45

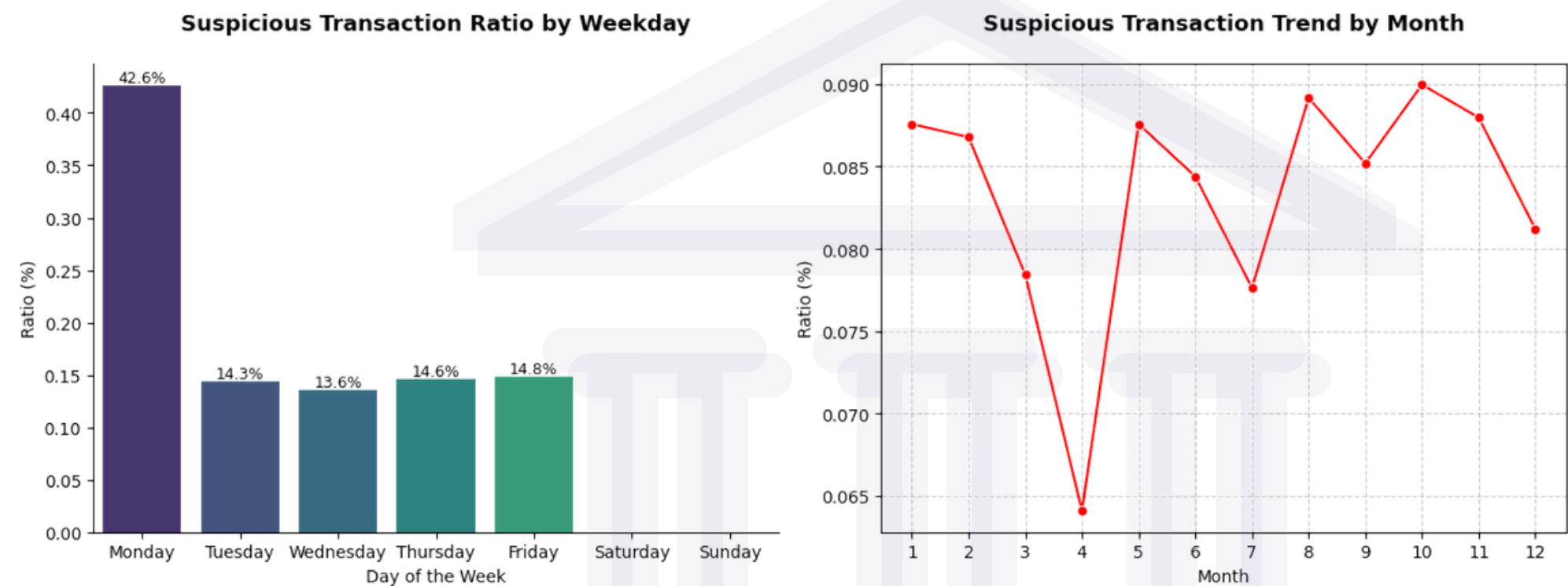
|   | IP Address     | CountAccount | TxCount |
|---|----------------|--------------|---------|
| 0 | 87.50.72.69    | 4            | 4       |
| 1 | 77.7.223.184   | 3            | 3       |
| 2 | 62.152.134.92  | 3            | 3       |
| 3 | 90.111.245.138 | 3            | 3       |
| 4 | 166.79.195.157 | 3            | 3       |

Number device shared by multiple accounts : 37

|   | DeviceID | CountAccount | TxCount |
|---|----------|--------------|---------|
| 0 | D000015  | 3            | 3       |
| 1 | D000673  | 3            | 3       |
| 2 | D000645  | 3            | 3       |
| 3 | D000294  | 3            | 3       |
| 4 | D000072  | 3            | 3       |

**Answer**  
The sharing of IPs and devices across multiple accounts is a **strong red flag** indicating potential **account farming, money laundering, or organized fraudulent behavior**.

## Are there or weekday trends for suspicious transactions?



### Answer

#### Trends in Suspicious Transactions

##### Weekday Pattern

- Highest on **Monday (≈42.6%)**, much higher than other days.
  - **Tue–Fri** stable around 14–15%.
  - **Weekend** nearly no suspicious activity.
- Likely due to **post-weekend backlog** or system activity resuming on Monday.

##### Monthly Pattern

- Overall stable (6–9%) with a **dip in April** and **peak around Aug–Oct**.
- Suggests **seasonal spending behavior** or **business cycle effects**.

### Insight

Suspicious transactions concentrate **early in the week** and **late in the year**, hinting at **operational or behavioral seasonality**.

# EDA – Business Question

Which merchants or regions have the highest concentration of large or frequent transactions?

💰 Amount threshold (95th percentile): 878.18  
🕒 Duration threshold (95th percentile): 265.00  
Number of frequent transactions (within 24 hours): 2  
🏪 Total merchants with frequent large transactions: 2

|   | MerchantID | TxCount | AvgAmount | MaxAmount |
|---|------------|---------|-----------|-----------|
| 0 | M025       | 1       | 146.67    | 146.67    |
| 1 | M038       | 1       | 474.77    | 474.77    |

## Answer

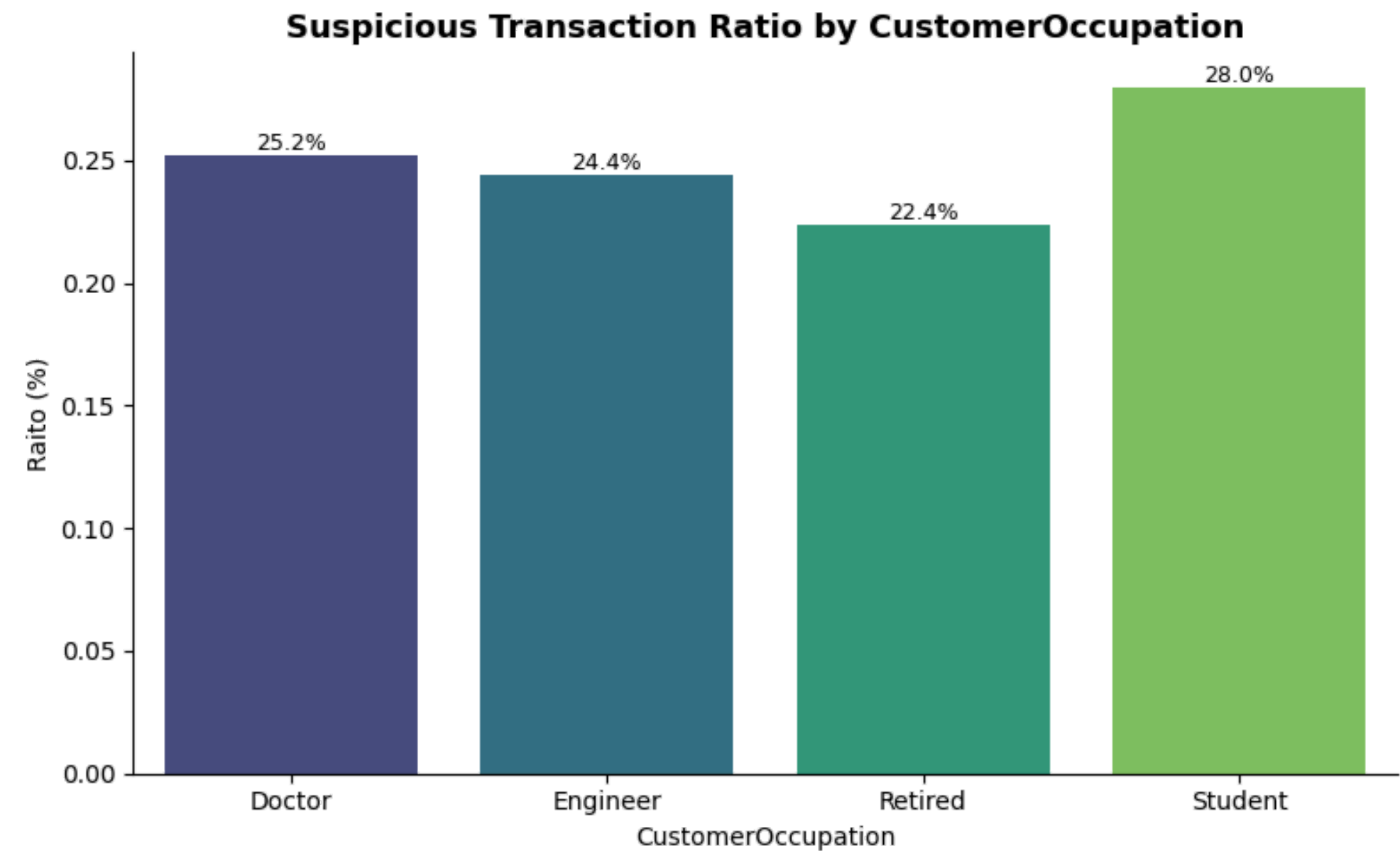
- When analyzing high-value transactions (Top 5%) within a 24-hour window, only **2 merchants (M025, M038)** were identified.
- However, each merchant performed **only one transaction**, meaning **no repeated transaction clusters** were detected.

👉 There is **no clear indication of abnormal or automated** behavior among merchants. High-value transactions appear to be **sporadic and time-independent**, not concentrated within short intervals.

# EDA – Business Question

Which occupations are linked to unusual transaction volumes or times?

💰 Amount threshold (95th percentile): 878.18  
🕒 Duration threshold (95th percentile): 265.00



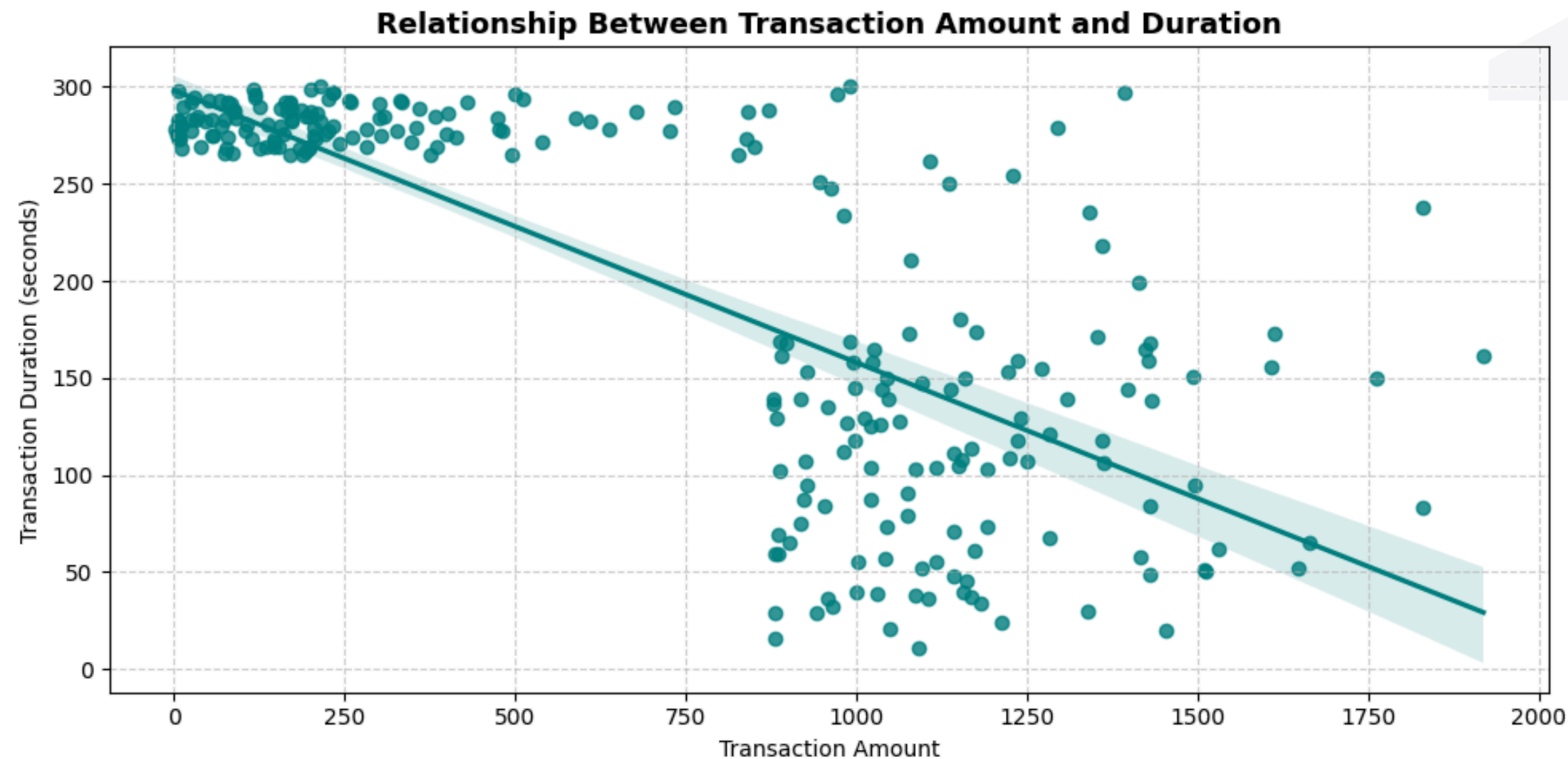
## Answer

- **Students** and **Doctors** account for over half of anomalous transactions, suggesting possible high activity or irregular spending patterns compared to other groups.

# EDA – Business Question

Is there a correlation between high transaction amount and long transaction duration (possible fraud)?

🔥 Amount threshold (95th percentile): 878.18  
🕒 Duration threshold (95th percentile): 265.00  
📊 Correlation between Transaction Amount and Duration:  $-0.767$



## Answer

- Strong negative correlation ( $r = -0.77$ ): higher-value transactions are processed faster.
- May indicate **automated approvals or missing verification checks** for large transactions.

# Data Preprocessing

## Feature Engineering

### TimeSinceLastTx

☞ Time (in seconds) since the previous transaction for the same **AccountID**. Detects **unusually frequent transactions**.

### Hour

☞ Hour of the day (0–23) — captures **daily activity patterns** (day vs. night).

### Month

☞ Transaction month (1–12) — used for **seasonal or monthly trends**.

### DayOfWeek

☞ Day of the week (1–7) — reflects **weekly transaction cycles**.

### UnixTime

☞ Timestamp encoded as **seconds since epoch (01/01/1970)** — preserves **temporal sequence**.

### Hour\_sin

☞ Cyclical encoding of **Hour** using  **$\sin(2\pi \cdot \text{hour}/24)$**  — models 24h periodicity.

### Hour\_cos

☞ Complementary cosine encoding of **Hour** — completes **daily cycle representation**.

### Day\_sin

☞ Cyclical encoding of **DayOfWeek** using  **$\sin(2\pi \cdot \text{day}/7)$**  — captures **weekly rhythm**.

### Day\_cos

☞ Complementary cosine encoding of **DayOfWeek** — ensures **smooth weekly continuity**.

Data columns (total 23 columns):

| #                                                    | Column              | Non-Null | Count    | Dtype    |
|------------------------------------------------------|---------------------|----------|----------|----------|
| ---                                                  | -----               | -----    | -----    | -----    |
| 0                                                    | TransactionID       | 2512     | non-null | category |
| 1                                                    | AccountID           | 2512     | non-null | category |
| 2                                                    | TransactionAmount   | 2512     | non-null | float64  |
| 3                                                    | TransactionType     | 2512     | non-null | category |
| 4                                                    | DeviceID            | 2512     | non-null | category |
| 5                                                    | IP Address          | 2512     | non-null | category |
| 6                                                    | MerchantID          | 2512     | non-null | category |
| 7                                                    | Channel             | 2512     | non-null | category |
| 8                                                    | CustomerAge         | 2512     | non-null | int64    |
| 9                                                    | CustomerOccupation  | 2512     | non-null | category |
| 10                                                   | TransactionDuration | 2512     | non-null | int64    |
| 11                                                   | LoginAttempts       | 2512     | non-null | category |
| 12                                                   | AccountBalance      | 2512     | non-null | float64  |
| 13                                                   | Region              | 2453     | non-null | category |
| 14                                                   | TimeSinceLastTx     | 2512     | non-null | float64  |
| 15                                                   | Hour                | 2512     | non-null | int32    |
| 16                                                   | Month               | 2512     | non-null | int32    |
| 17                                                   | DayOfWeek           | 2512     | non-null | int32    |
| 18                                                   | UnixTime            | 2512     | non-null | int64    |
| 19                                                   | Hour_sin            | 2512     | non-null | float64  |
| 20                                                   | Hour_cos            | 2512     | non-null | float64  |
| 21                                                   | Day_sin             | 2512     | non-null | float64  |
| 22                                                   | Day_cos             | 2512     | non-null | float64  |
| dtypes: category(10), float64(7), int32(3), int64(3) |                     |          |          |          |
| memory usage: 431.8 KB                               |                     |          |          |          |

# Data Preprocessing

## Handling Skewness

 Skewness for dataset:

| Feature             | Skewness  | Remark                  |
|---------------------|-----------|-------------------------|
| TransactionAmount   | +1.740107 | Highly skewed           |
| TimeSinceLastTx     | +1.732880 | Highly skewed           |
| Hour_cos            | +0.764228 | Moderately skewed       |
| Hour                | +0.711157 | Moderately skewed       |
| AccountBalance      | +0.601201 | Moderately skewed       |
| TransactionDuration | +0.599432 | Moderately skewed       |
| Hour_sin            | -0.214948 | Approximately symmetric |
| CustomerAge         | +0.147864 | Approximately symmetric |
| Day_sin             | -0.072487 | Approximately symmetric |
| Month               | -0.056899 | Approximately symmetric |
| UnixTime            | -0.052795 | Approximately symmetric |
| Day_cos             | -0.034139 | Approximately symmetric |
| DayOfWeek           | -0.006695 | Approximately symmetric |



 Skewness for dataset:

| Feature                | Skewness  | Remark                  |
|------------------------|-----------|-------------------------|
| Hour_cos               | +0.764228 | Moderately skewed       |
| PT_TimeSinceLastTx     | -0.746342 | Moderately skewed       |
| Hour                   | +0.711157 | Moderately skewed       |
| Hour_sin               | -0.214948 | Approximately symmetric |
| PT_AccountBalance      | -0.162008 | Approximately symmetric |
| CustomerAge            | +0.147864 | Approximately symmetric |
| PT_TransactionDuration | -0.078108 | Approximately symmetric |
| Day_sin                | -0.072487 | Approximately symmetric |
| Month                  | -0.056899 | Approximately symmetric |
| UnixTime               | -0.052795 | Approximately symmetric |
| PT_TransactionAmount   | -0.051555 | Approximately symmetric |
| Day_cos                | -0.034139 | Approximately symmetric |
| DayOfWeek              | -0.006695 | Approximately symmetric |

### ✔ Insight

- Before transformation, several features (e.g., **TransactionAmount**, **TimeSinceLastTx**, **AccountBalance**) showed **highly skewed**, indicating **long-tailed distributions** caused by a few extreme values.
- After applying **power transformation**, most variables became **approximately symmetric**, reducing bias and improving model performance.
- Only **Hour\_cos** and **PT\_TimeSinceLastTx** remain **moderately skewed**, which is acceptable for unsupervised models like **K-Means** or **Isolation Forest**,...

## Re-check Outliers

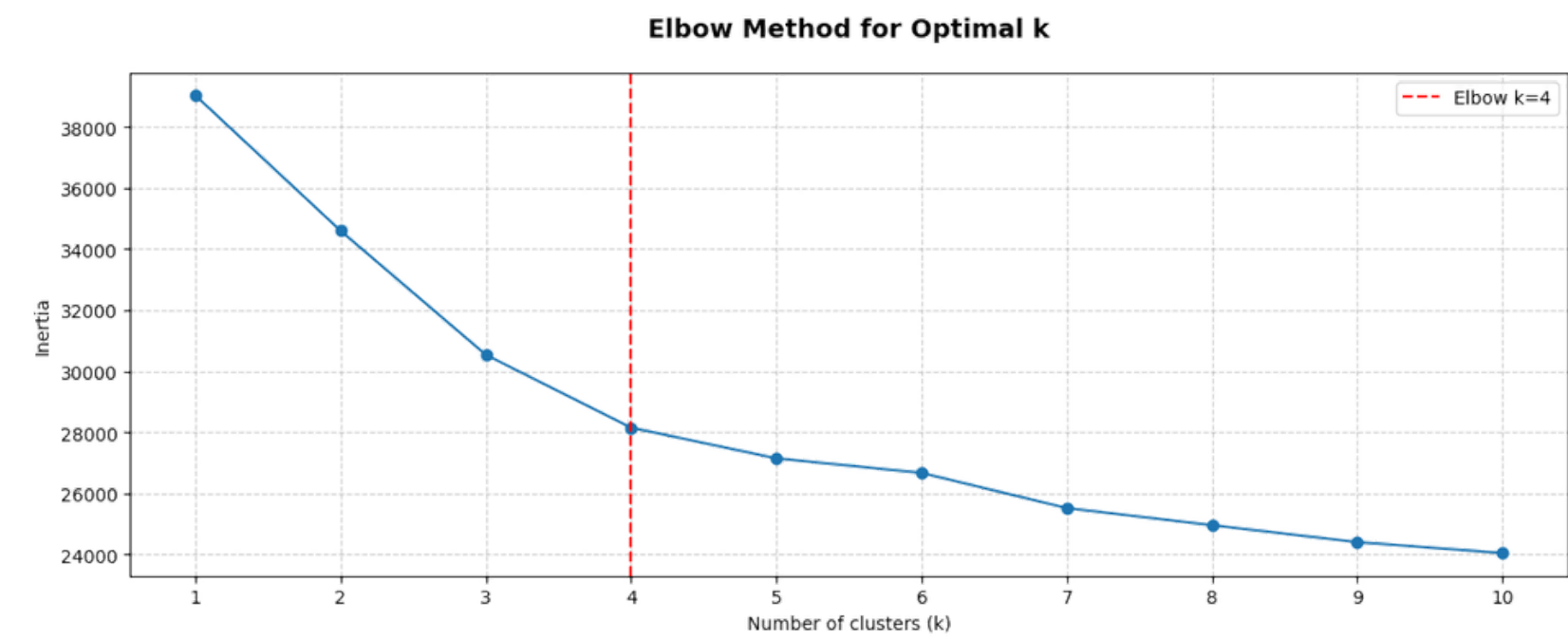
```
=====
 Data Checking outlier
=====
✔ No outliers detected in the selected features.
```

## Scale and Encode

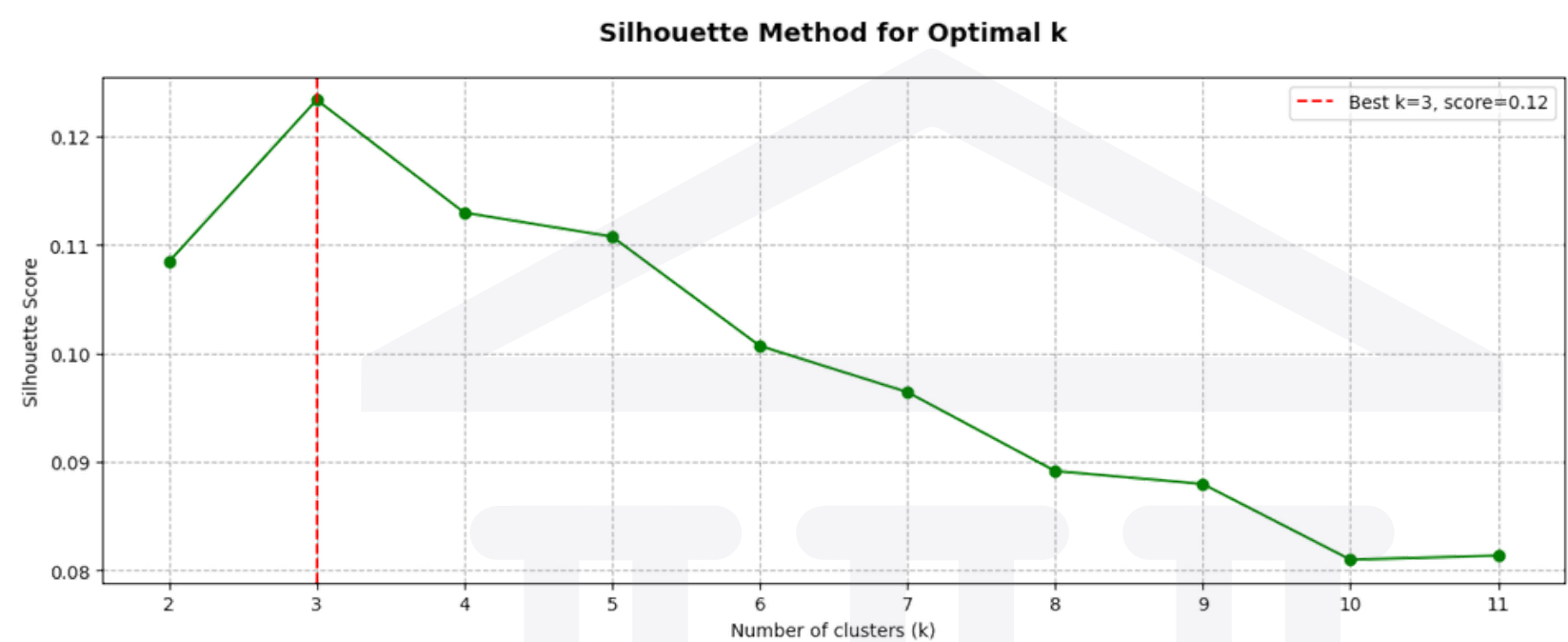
- Imputation:** Median (numeric), Most frequent (categorical).
- Scaling:** StandardScaler for all numerical features.
- Encoding:** OneHot for categorical variables.
- Output:** Unified **preprocessor** → clean, consistent dataset for ML.

✔ **Insight:** Preprocessing ensures fair scaling & robust handling of features.

# Modeling – KMEAN



Optimal k suggested by Elbow: 4



Optimal k suggested by Silhouette: 3

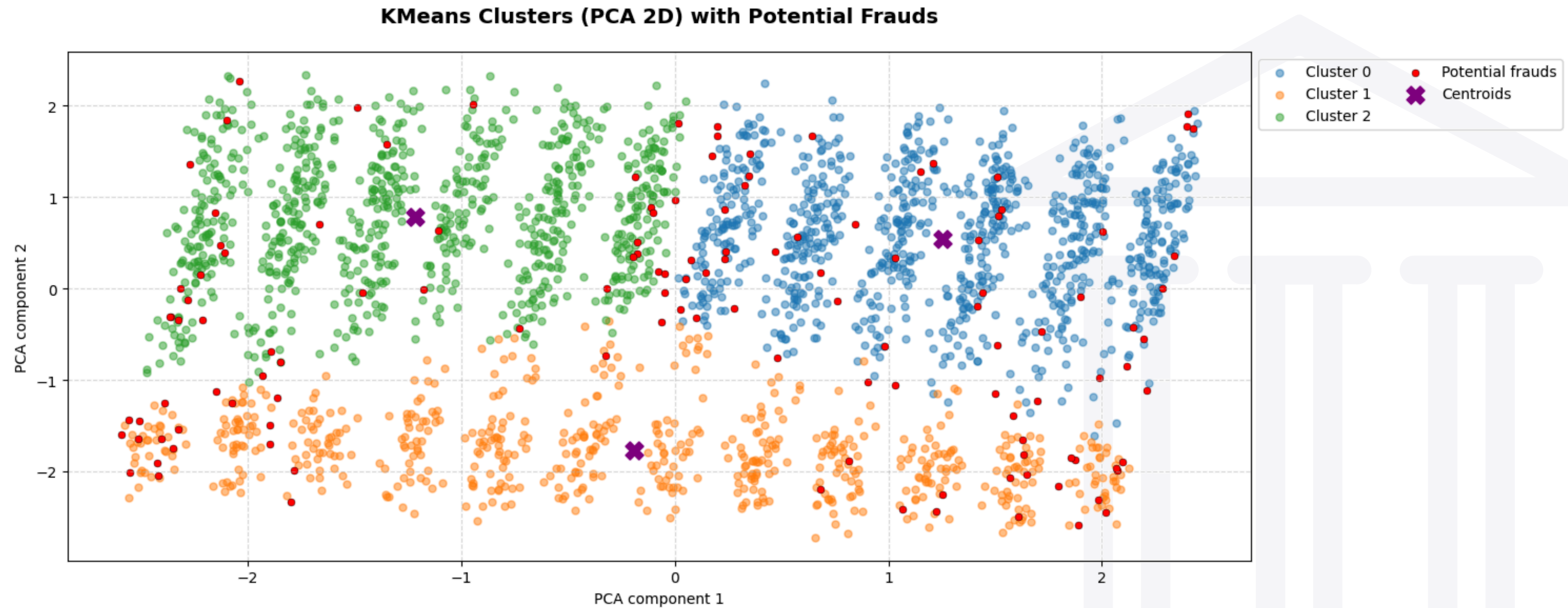
Number of potential frauds detected: 126

|   | TransactionID | AccountID | TransactionAmount | TransactionType | DeviceID | IP Address    | MerchantID | Channel |
|---|---------------|-----------|-------------------|-----------------|----------|---------------|------------|---------|
| 0 | TX002326      | AC00003   | 20.17             | Debit           | D000287  | 221.23.158.87 | M088       | ATM     |
| 1 | TX001456      | AC00015   | 774.48            | Debit           | D000233  | 76.136.92.235 | M013       | Branch  |
| 2 | TX001498      | AC00018   | 1228.81           | Debit           | D000381  | 63.191.171.41 | M034       | ATM     |
| 3 | TX001821      | AC00021   | 263.59            | Debit           | D000333  | 150.25.200.56 | M052       | Branch  |
| 4 | TX002112      | AC00021   | 762.74            | Debit           | D000196  | 8.136.178.116 | M053       | ATM     |

✓ The **Elbow Method** suggested **k = 4**, while the **Silhouette Score** peaked at **k = 3** → choosing **k = 3** provides a good balance between **cluster separation** and **model stability**.

✓ With **k = 3**, the model identified **three main behavioral clusters**, including **126 transactions flagged as potential frauds**.

# Modeling – KMEAN

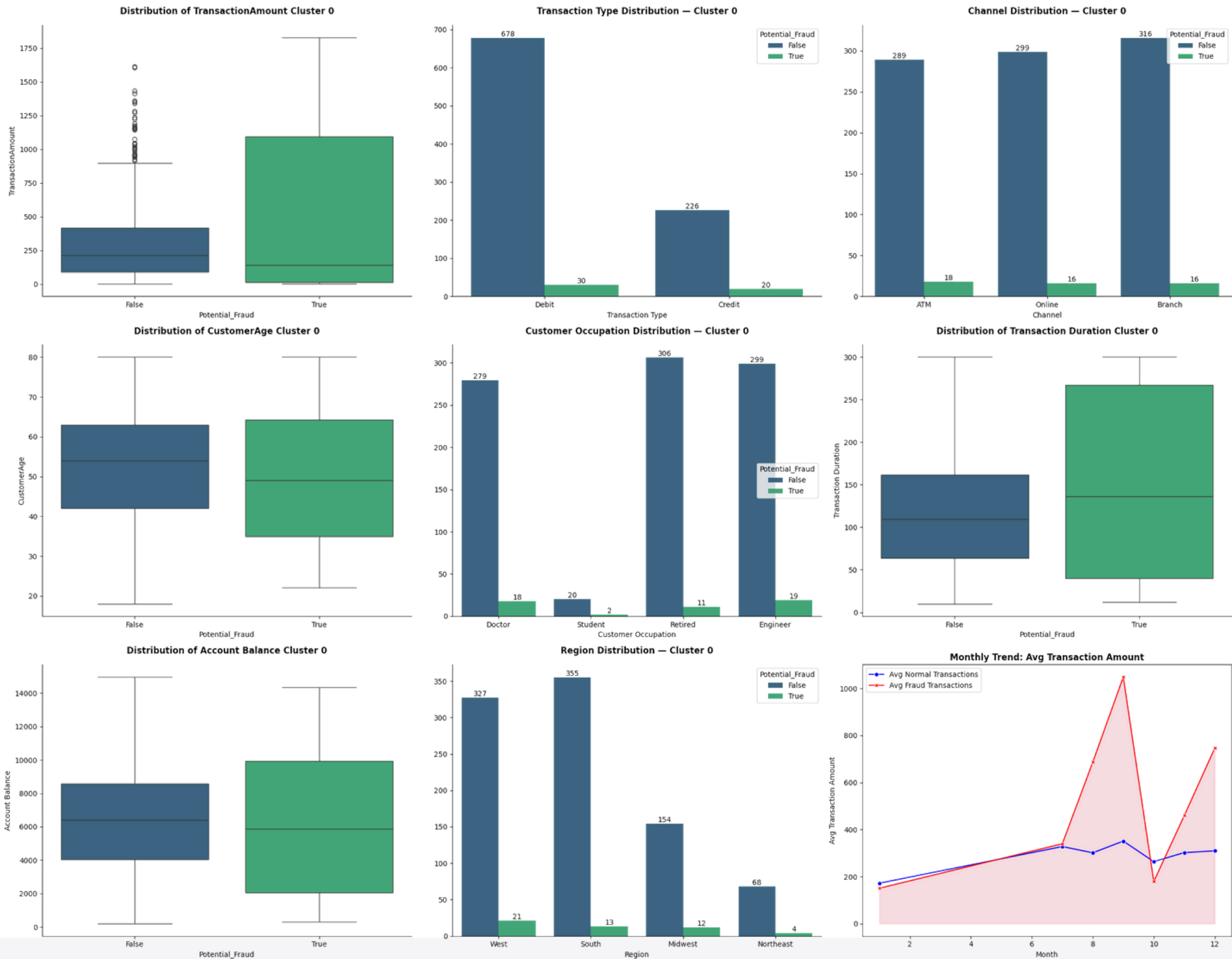


## ◆ K-Means Clustering Summary

- The K-Means model ( $k = 3$ ) forms **three distinct behavioral clusters**, well separated in PCA space.
- **126 transactions ( $\approx$  top 5%)** were flagged as potential frauds based on distance from cluster centroids.
- ☒ Overall, K-Means effectively **captures transaction structure and isolates anomalous behaviors** for fraud risk assessment.

# Modeling – KMEAN

Cluster 0: Fraud Behavior across Multiple Features and Monthly Trend



## Cluster 0 – Fraud Behavior Overview

- **Most transactions are legitimate; fraud is a small share (~5%).**
- **Fraud = higher amount & longer duration.**
- **Within fraud:** Debit = 60% (30/50); Online  $\approx$  Branch < ATM.
- **Age & occupation:** slightly older customers; **Doctors** and **Engineers** appear more often.
- **Higher account balances** among fraud cases.
- Regional share: higher in **West and South**; strong volatility, **peaking around September–October.**

# Modeling – KMEAN

Cluster 1: Fraud Behavior across Multiple Features and Monthly Trend



## Cluster 1 – Fraud Behavior Overview

Most transactions are legitimate; fraud is a small share (~5%).  
Fraud = higher amount & longer duration.  
Within fraud: Debit  $\approx$  69% (29/42); Online > Branch > ATM.  
Age & occupation: mainly **younger users**, especially **Students**.  
Lower account balances compared to Cluster 0.  
Regional share: concentrated in **Midwest and West**; fraud peaks around **Aug–Nov**.

# Modeling – KMEAN

Cluster 2: Fraud Behavior across Multiple Features and Monthly Trend



## \$ Cluster 2 – Fraud Behavior Overview

- Most transactions are legitimate; fraud is a small share (~5%).
- Fraud = higher amount & longer duration.
- Within fraud: Debit  $\approx$  76% (26/34); Online  $\approx$  Branch  $\triangleright$  ATM.
- Age & occupation: mainly middle-aged customers, especially Retired and Doctor groups.
- Higher account balances among fraud cases.
- Regional share: concentrated in West and South; fraud fluctuates strongly around March–May.

## 👛 Cluster 0 – Stable High-Value Users

- High-value, consistent transactions with low fraud risk.
- Fraud mainly from older professionals (Doctors, Engineers) in West/South; peaks Sep–Oct.

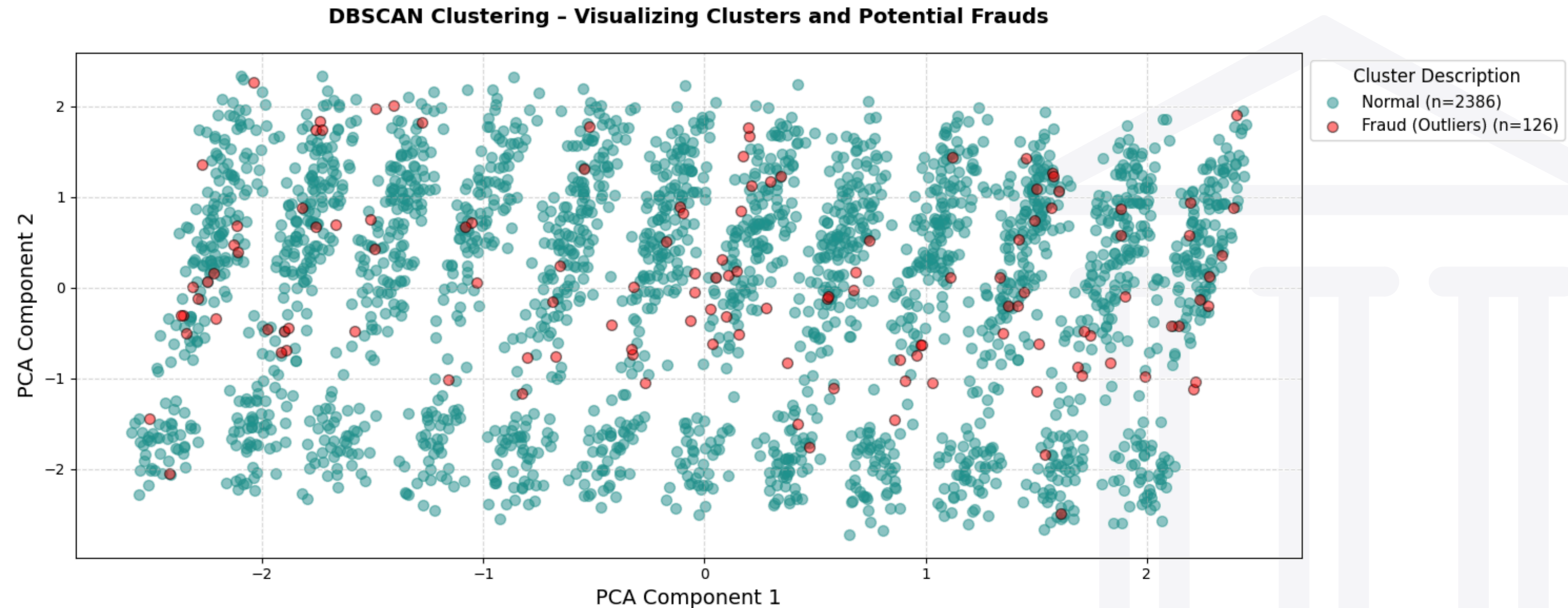
## 🏠 Cluster 1 – Young Online Users

- Moderate spenders, mostly young users (Students) using Debit via Online channels.
- Lower balances, fraud spikes Aug–Nov in Midwest/West.

## \$ Cluster 2 – Activity Mature Users

- Frequent, high-value transactions; middle-aged (Retired/Doctors) dominate.
- Higher balances; fraud concentrated in West/South, fluctuating Mar–May.

# Modeling – DBSCAN



## ❖ DBSCAN Clustering Summary

- Number of potential frauds detected by DBSCAN: **126**.
- The chart illustrates the **DBSCAN clustering results** after dimensionality reduction using **PCA (2D)**.
- The algorithm successfully identified **one main cluster (Normal)**, and several **outliers (Fraud / Outliers)**.

## Insight:

- DBSCAN effectively detects **density-based outliers**, clearly separating **normal and potentially fraudulent activities** without requiring a predefined number of clusters like Kmean.
- The suspicious or anomalous points account for roughly **5% of the dataset**, which aligns well with the typical fraud occurrence rate in real-world scenarios.

# Modeling – DBSCAN

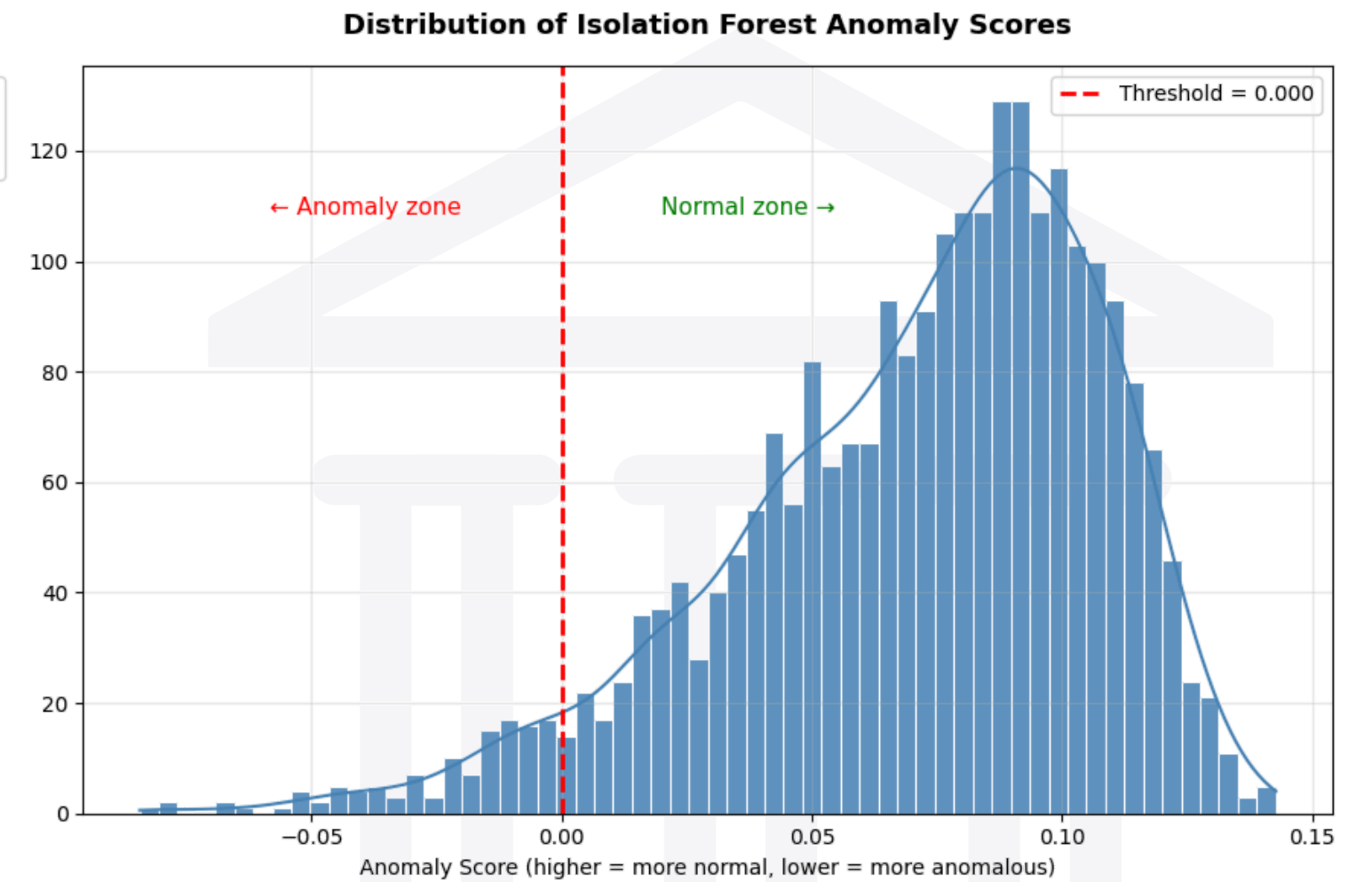
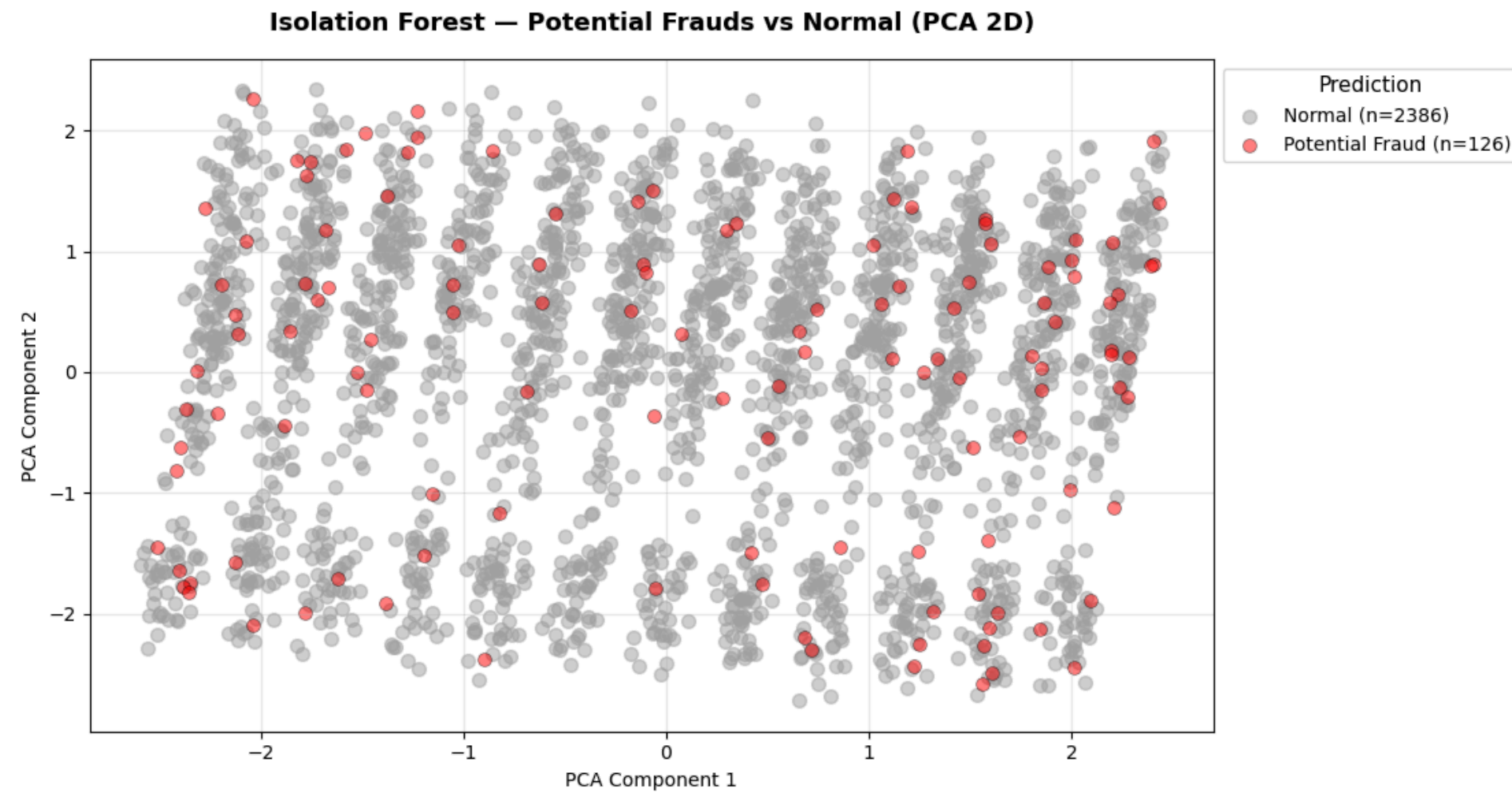
Fraud Behavior across Multiple Features and Monthly Trend



## 💡 Fraud Behavior Summary (DBSCAN Results)

- **Most transactions are legitimate; fraud is a small share (~5%).**
- **Fraud = higher amount & longer duration.**
- **Within fraud: Credit  $\approx$  50% (63/126); Online  $\succ$  Branch  $\succ$  ATM.**
- **Age & occupation:** no clear separation; slight middle-age tilt, **Engineers/Doctors** appear more often but not dominant.
- **Higher account balances** among fraud cases.
- **Regional share:** no clear separation; **strong month-to-month volatility.**

# Modeling – Isolation Forest



## 🧠 Isolation Forest — Fraud Detection Summary

- Isolation Forest detects about 5% of transactions as anomalies, located at the edges of main clusters.
- The model effectively separates **normal** and **potentially fraudulent** behaviors, making it suitable for initial screening.

# Modeling – Isolation Forest

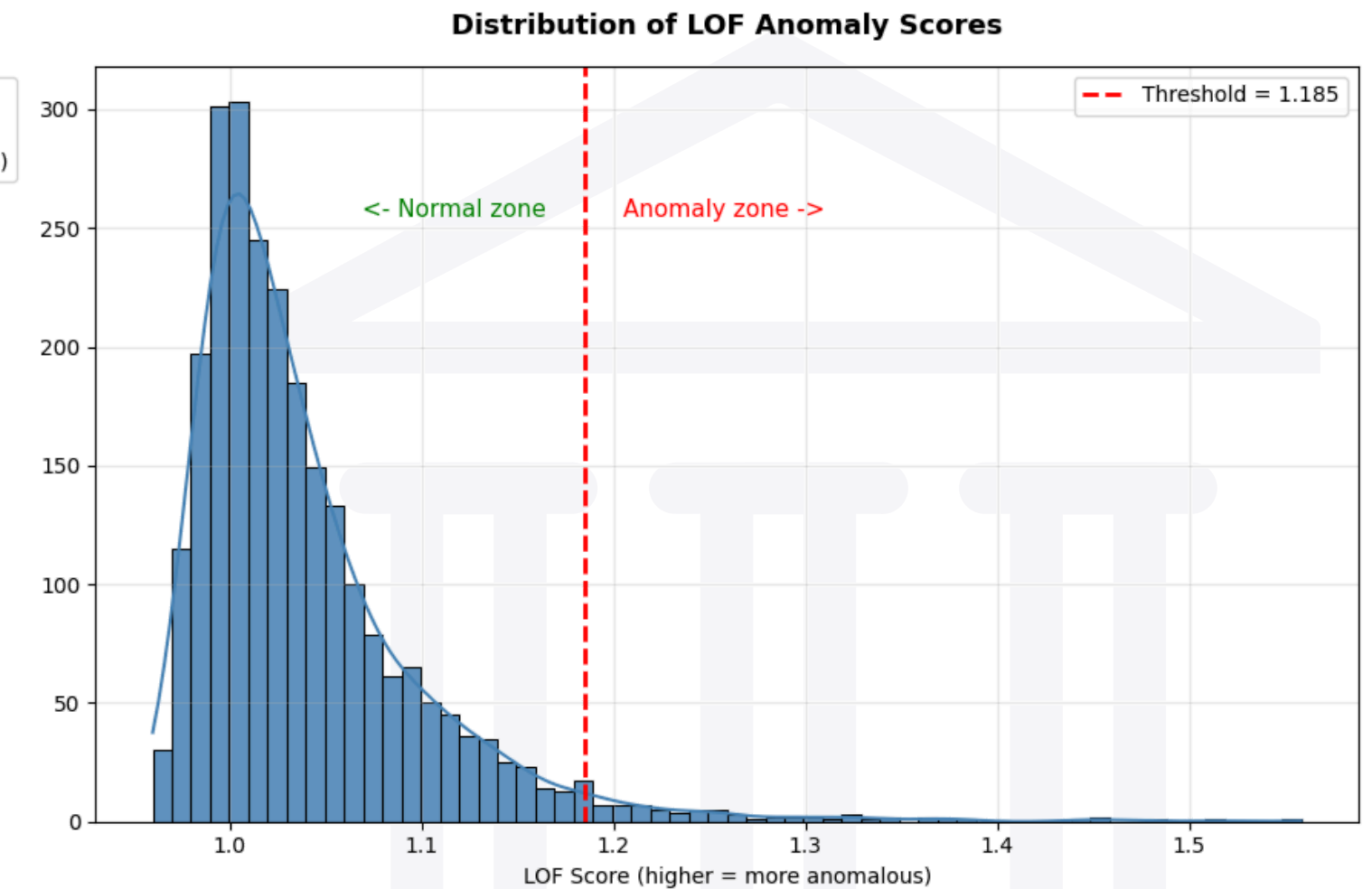
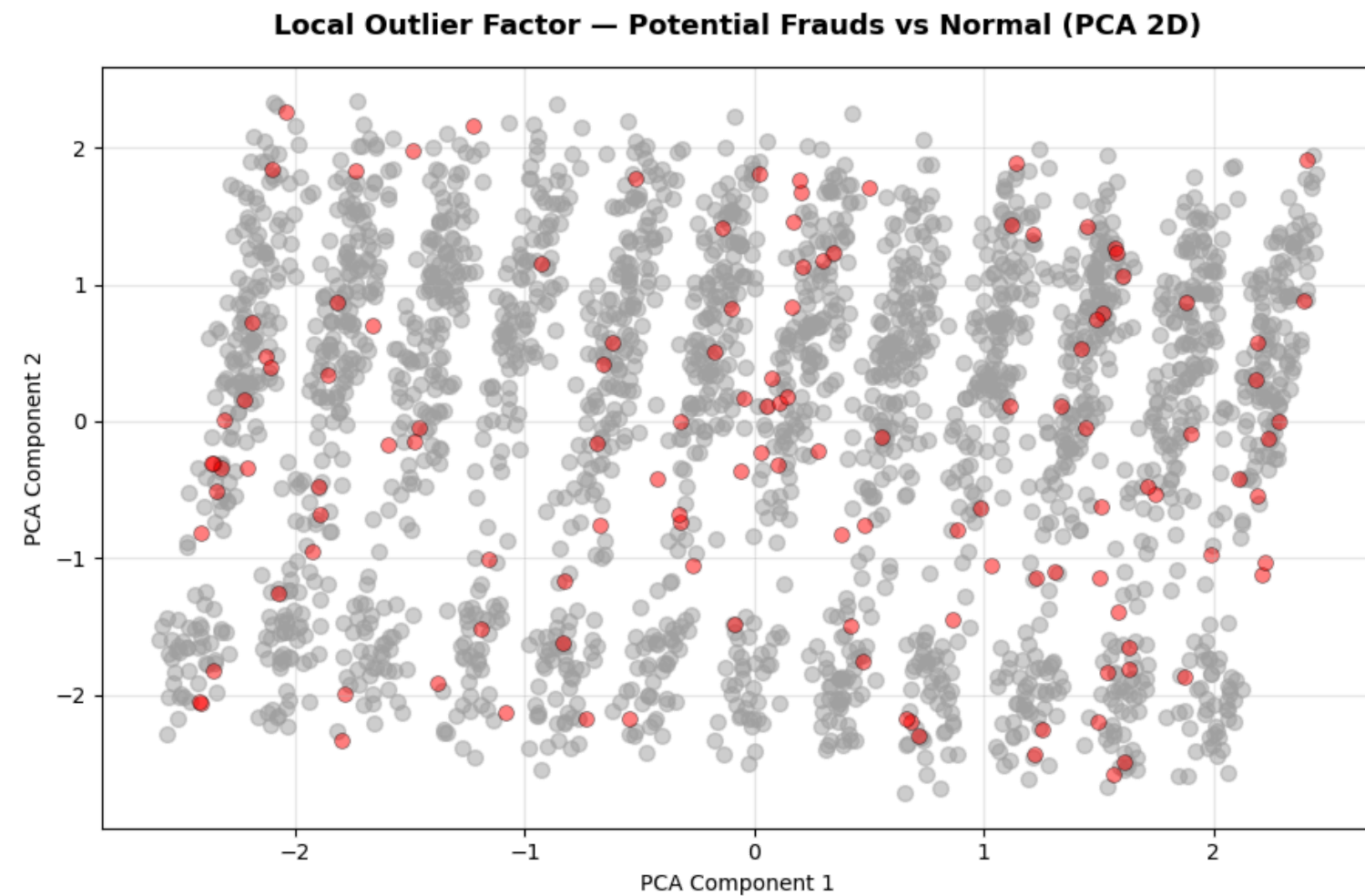
Fraud Behavior across Multiple Features and Monthly Trend



## 💡 Fraud Behavior Summary (Isolation Forest Results)

- Most transactions are legitimate; fraud is a small share.
- Fraud = **higher amount & longer duration**.
- Within fraud: **Credit ~61%; Online > Branch > ATM**.
- **Age & occupation:** no clear separation.
- **Higher account balances** among fraud cases.
- **Regional share:** relatively higher in **Midwest/Northeast**; **strong monthly volatility**.

# Modeling – Local Outlier Factor



## ✳ Local Outlier Factor (LOF) — Fraud Detection Summary

- Clear visual separation between **normal (gray)** and potential **fraud (red)** transactions.
- The histogram shows a **right-skewed distribution** with most transactions below the anomaly threshold (LOF = 1.092).
- LOF identifies about **5% of transactions** as potential frauds, effectively isolating **low-density, irregular behaviors** that deviate from normal transaction patterns.

# Modeling – Local Outlier Factor

Fraud Behavior across Multiple Features and Monthly Trend



## 💡 Fraud Behavior Summary (LOF Results)

- **Most transactions are legitimate; fraud is a small share (~5%).**
- **Fraud = higher amount & longer duration.**
- **Within fraud:** Credit  $\approx$  55% (47/85); Online > Branch > ATM.
- **Age & occupation:** slight middle-age tilt; Doctors and Engineers appear more frequently.
- **Higher account balances among fraud cases.**
- **Regional share:** relatively higher in West/South; strong monthly fluctuations, especially early and mid-year.

 **Overall Insight**

- ~**5%** transactions flagged as anomalous across models.
- Fraud = **higher amounts & longer durations**.
- **Within fraud: Debit dominates; Online** shows **higher exposure** (**Online > Branch > ATM**).
- **Regions:** fraud counts higher in **West/South**; **rate-wise** can be higher in **Midwest/Northeast**.
- **Seasonality:** fraud activity **peaks around Mar-May and Aug-Oct** (not June/December), indicating possible cyclical patterns.

# Conclusion

# Appendix – RFM Analysis

Customer segmentation based on RFM Score

| Segment                    | Description                                                                  | RFM Condition                  |
|----------------------------|------------------------------------------------------------------------------|--------------------------------|
| Champions / Best Customers | Most valuable customers: recent activity, frequent transactions, high spend. | $R \geq 4, F \geq 4, M \geq 4$ |
| Loyal Customers            | Loyal customers: purchase frequently and spend consistently well.            | $F \geq 4, M \geq 3$           |
| Potential Loyalists        | New customers with decent frequency/spending, potential to become loyal.     | $R \geq 4, F \geq 2$           |
| At Risk                    | Previously frequent/high spenders but inactive for a long time.              | $R \leq 2, F \geq 3$           |
| New Customers              | Recently joined, made transactions recently but with low frequency.          | $R \geq 4, F = 1$              |
| Hibernating / Churn        | Customers inactive for a long time with low spending.                        | $R = 1, F = 1, M = 1$          |
| Others                     | Customers not falling into the above categories.                             | Different from above rules     |

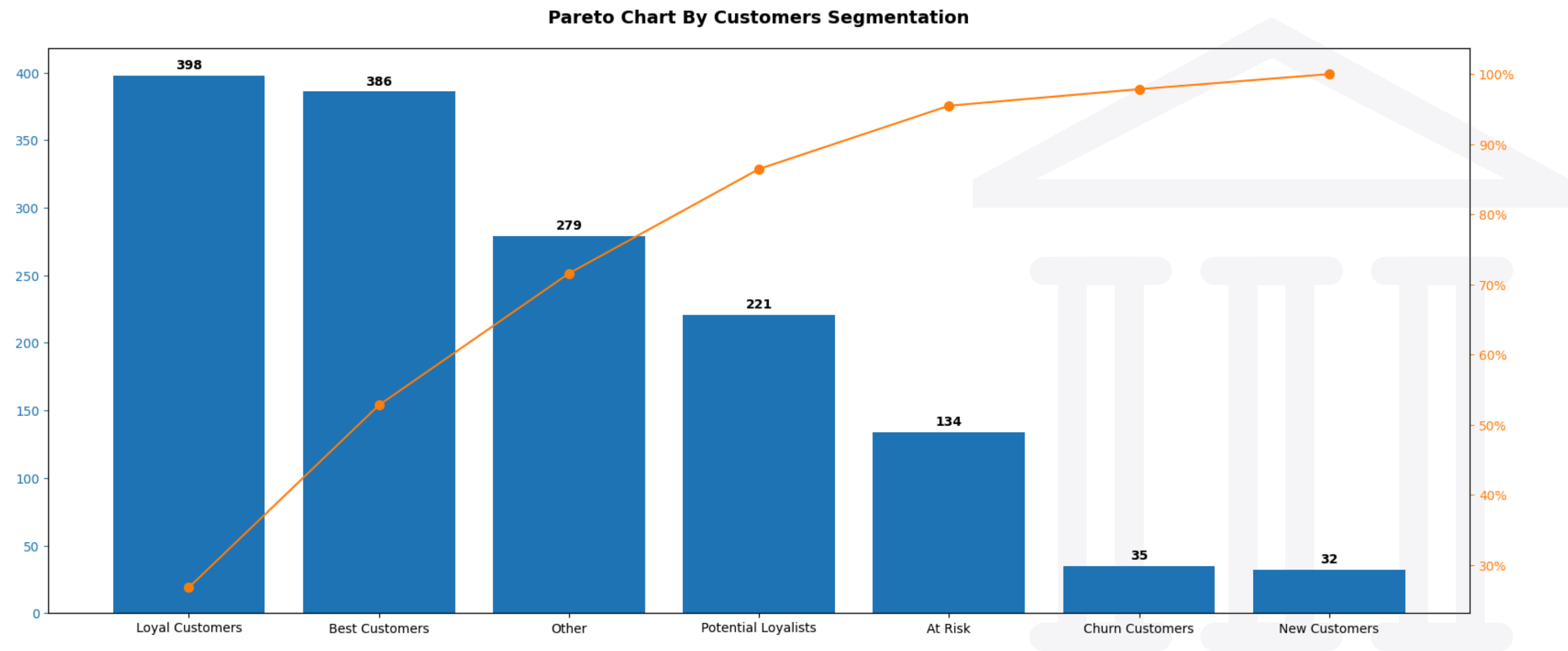
RFM Summary by Segment

| RFM_Segment         | Recency |     |     | Frequency |     |     | Monetary |      |      | AccountID |
|---------------------|---------|-----|-----|-----------|-----|-----|----------|------|------|-----------|
|                     | mean    | min | max | mean      | min | max | mean     | min  | max  | count     |
| Best Customers      | 19      | 1   | 36  | 8         | 6   | 12  | 2558     | 1639 | 5570 | 85        |
| Loyal Customers     | 62      | 1   | 188 | 7         | 6   | 11  | 2144     | 1128 | 4393 | 99        |
| At Risk             | 106     | 64  | 221 | 5         | 4   | 6   | 1291     | 384  | 2980 | 49        |
| Potential Loyalists | 19      | 1   | 36  | 5         | 3   | 8   | 1283     | 312  | 3213 | 82        |
| Other               | 94      | 38  | 273 | 3         | 1   | 7   | 1008     | 76   | 2642 | 125       |
| New Customers       | 14      | 4   | 36  | 2         | 1   | 3   | 645      | 78   | 1602 | 20        |
| Churn Customers     | 227     | 111 | 364 | 2         | 1   | 3   | 301      | 7    | 607  | 35        |

Overall Insight:

- **Best & Loyal Customers:** though fewer in number, they generate the highest value → priority for retention and upselling.
- **At Risk & Churn Customers:** warning signs of attrition → need targeted win-back campaigns.
- **Potential Loyalists & New Customers:** represent **growth opportunities** → with the right nurturing, they can transition into Loyal or Best Customers.

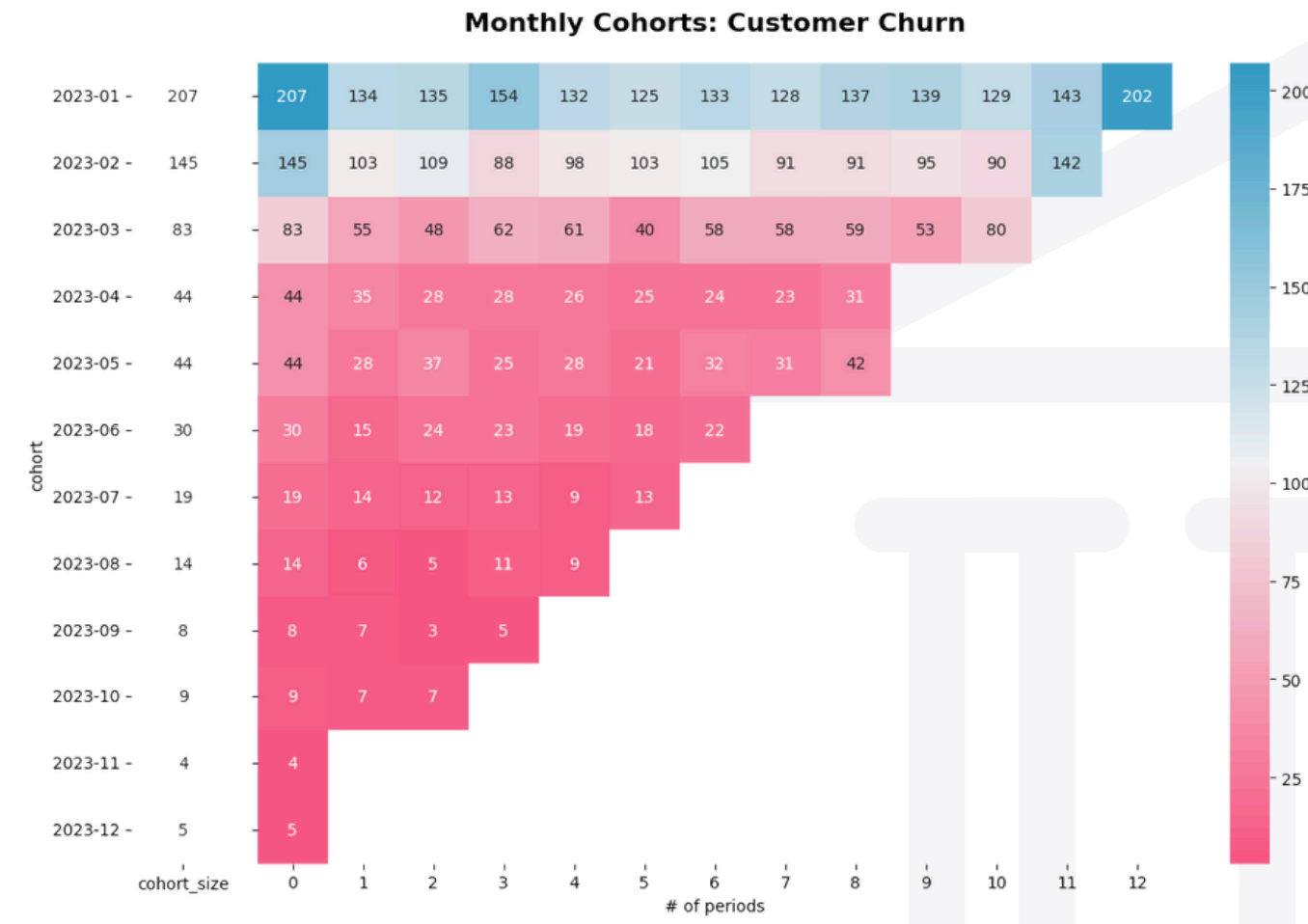
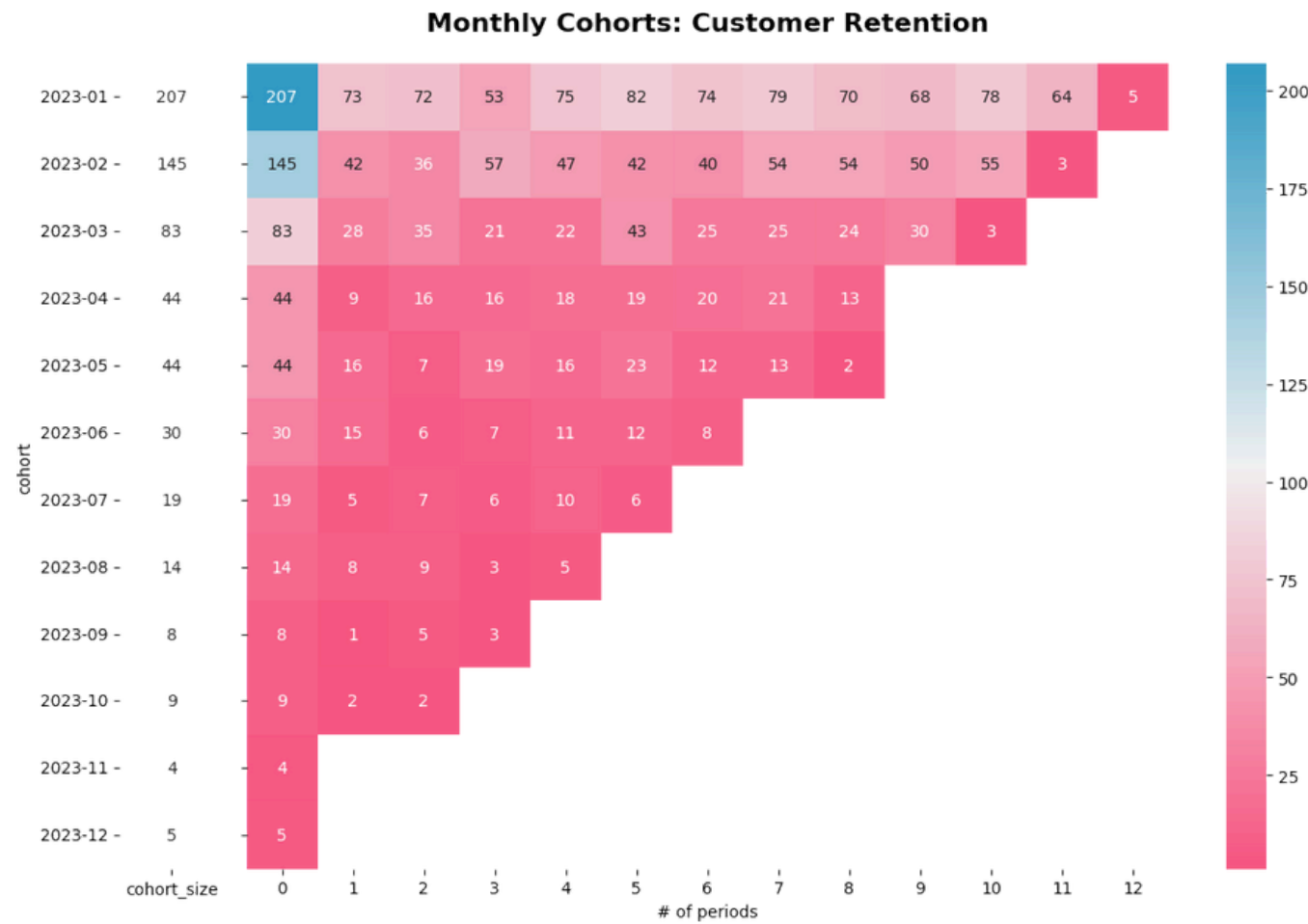
# Appendix – RFM Analysis



## Summary

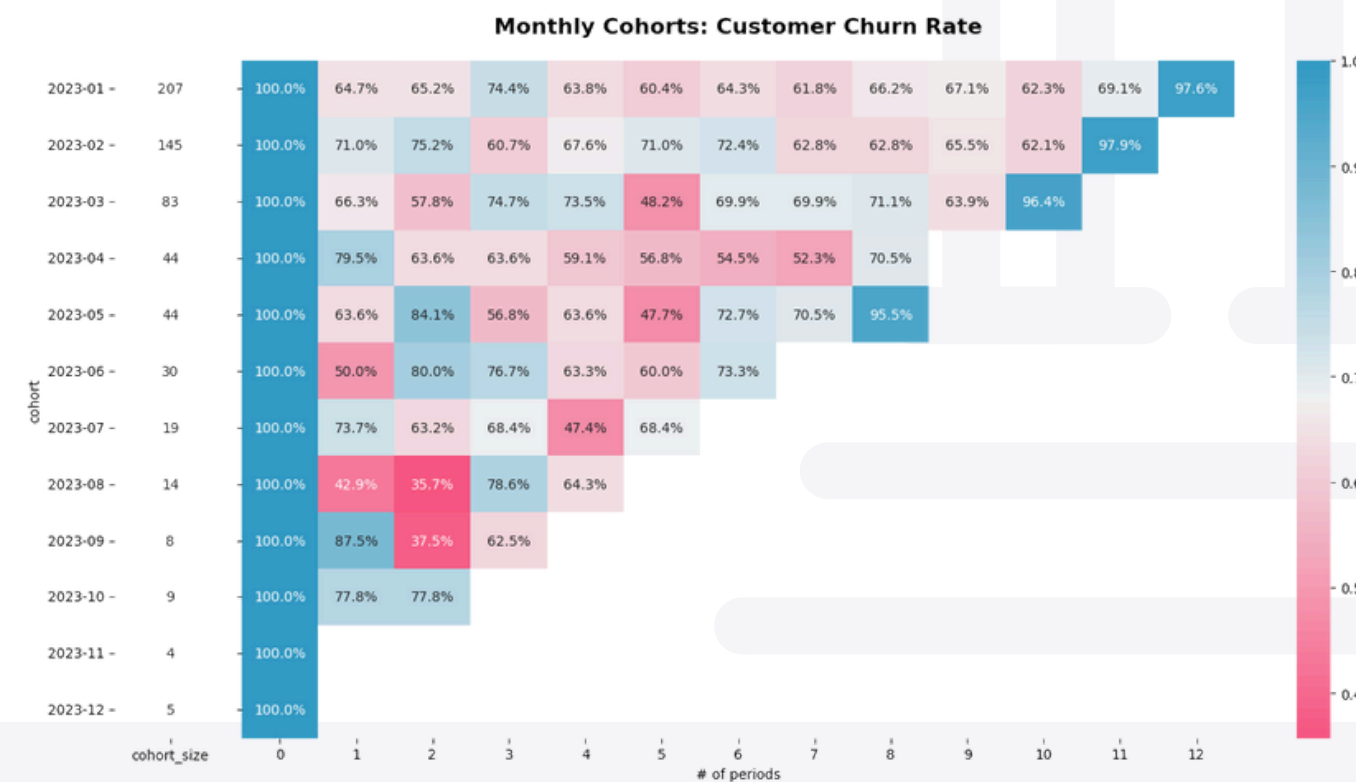
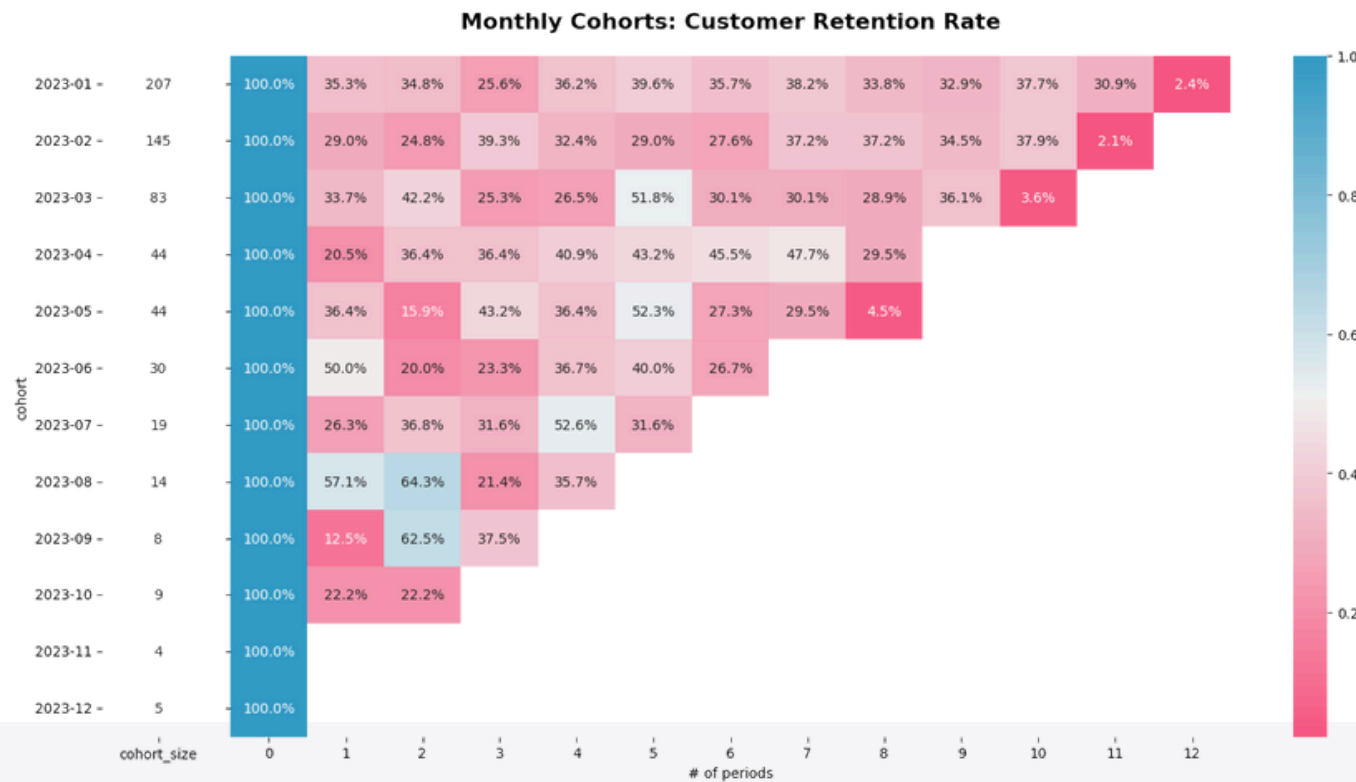
The chart demonstrates that while the **bulk of customers are concentrated in Loyal, Best, and Other groups**, smaller segments like **Potential Loyalists, At Risk, and New Customers** hold **strategic value**. They can either be converted into high-value customers or reactivated to reduce churn risk.

## Appendix – Cohort Analysis

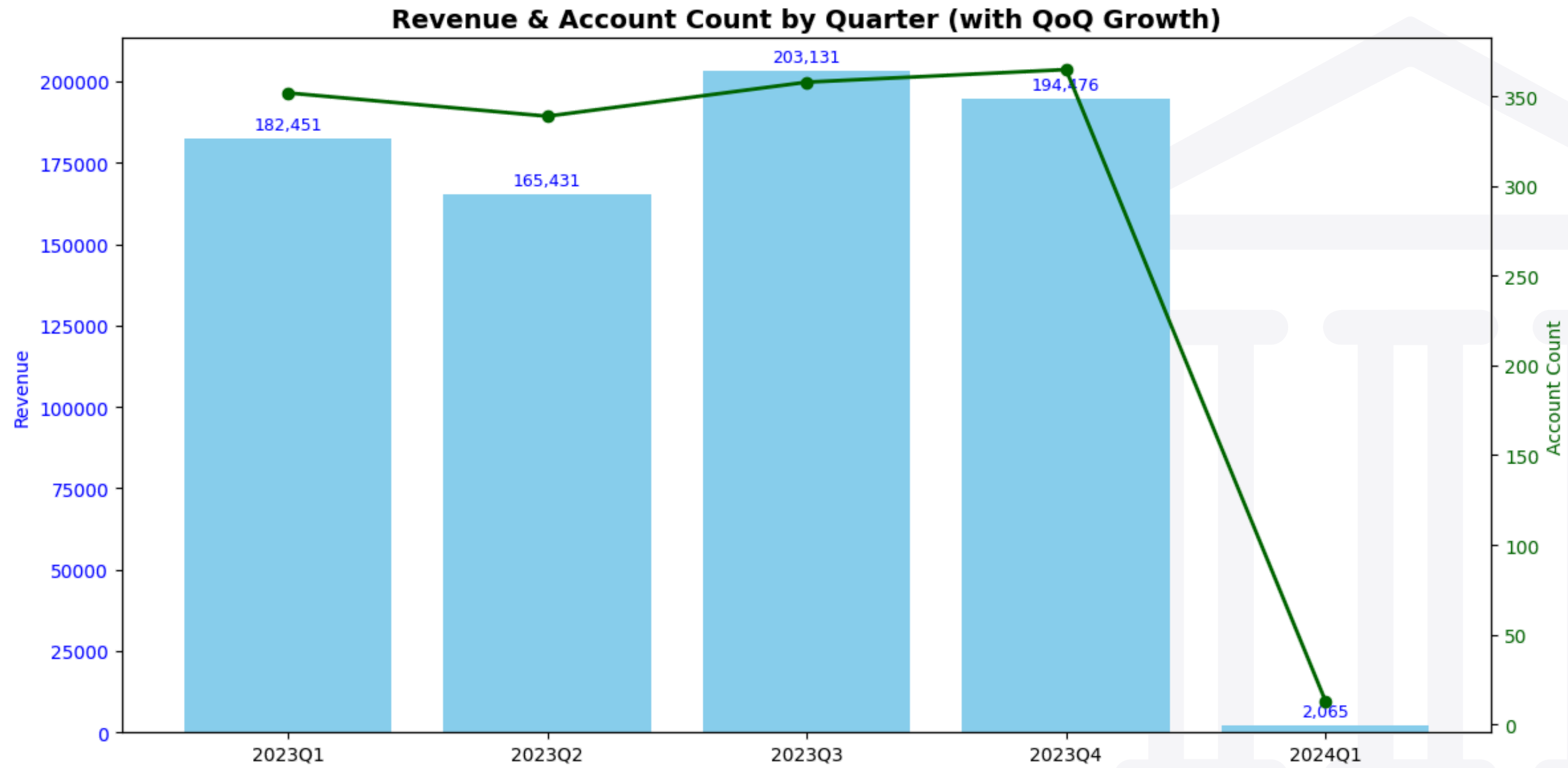


## In summary:

- Cohort analysis shows that **most customers make only 1–2 transactions.**
- **Retention declines rapidly, churn rises sharply from the first month.**
- Improving performance requires **customer retention strategies:** after-sales engagement, repurchase incentives, and enhancing the overall customer journey.



# Appendix – QoQ Analysis



## In summary:

👉 In summary: 2023 was stable with positive growth in Q3, but **2024Q1 anomalies** indicate a need to validate the dataset.

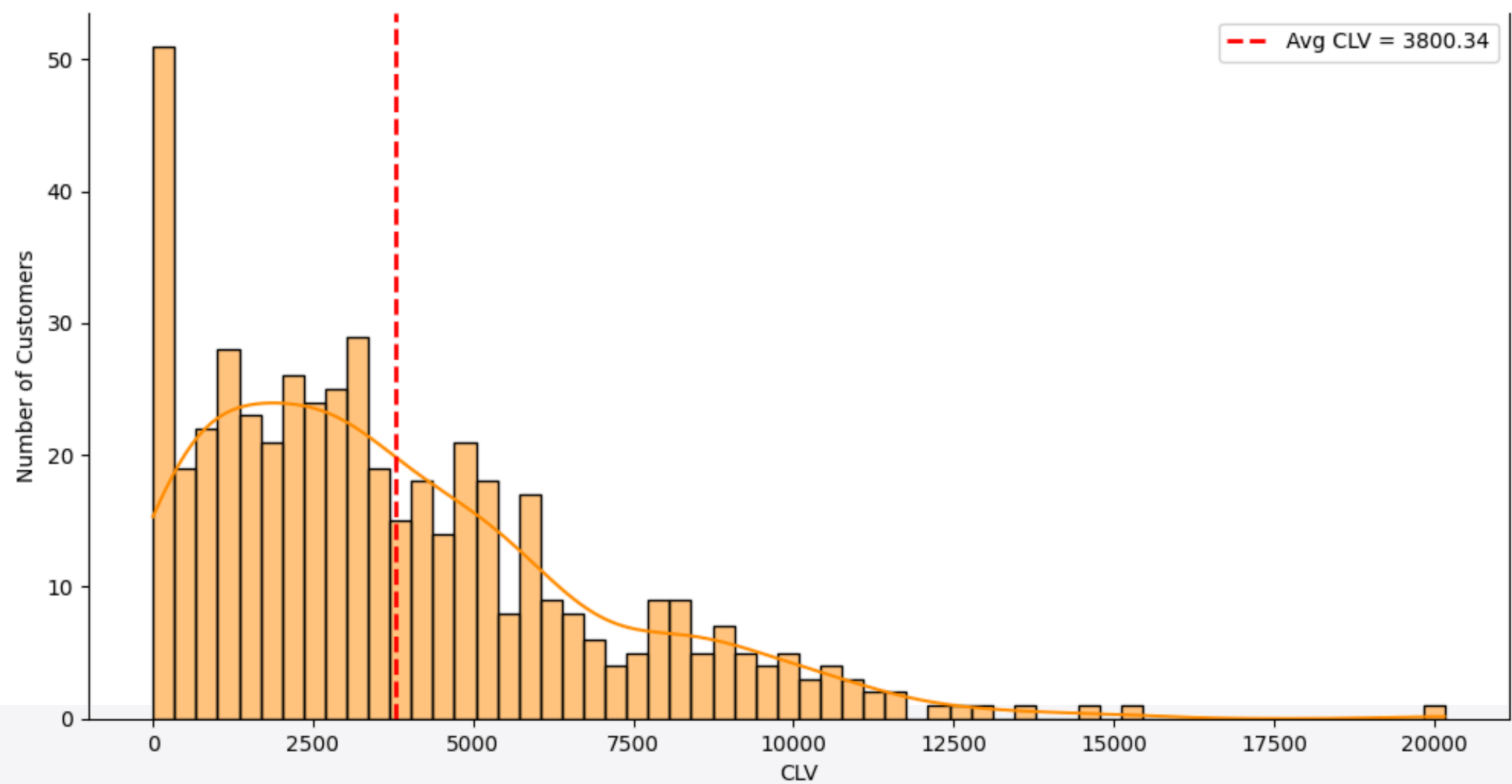
# Appendix – CLV Analysis

CLV = Total profit a customer generates during their relationship with the company. It measures **long-term customer value** and helps optimize **marketing spend, retention, and growth strategy**.

$$CLV = (AveragePurchaseValue \times GrossMargin) \times PurchaseFrequency \times Lifespan$$

- **Average Purchase Value × Gross Margin:** Profit per purchase
- **Purchase Frequency:** How often customers buy (e.g., times per year)
- **Lifespan:** Average years a customer stays active
- **Interpretation:** Total lifetime profit from an average customer.

Distribution of Customer Lifetime Value (CLV)

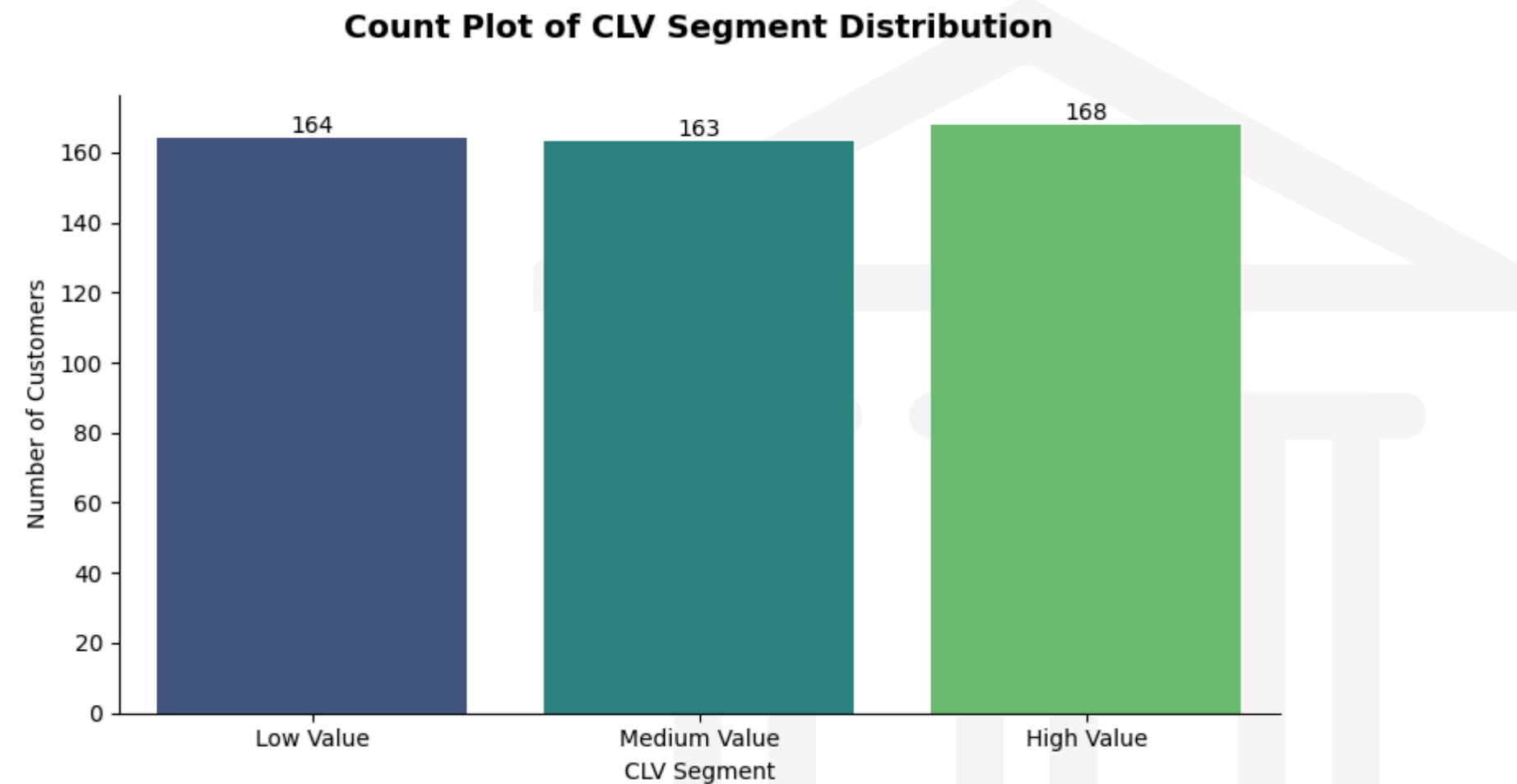
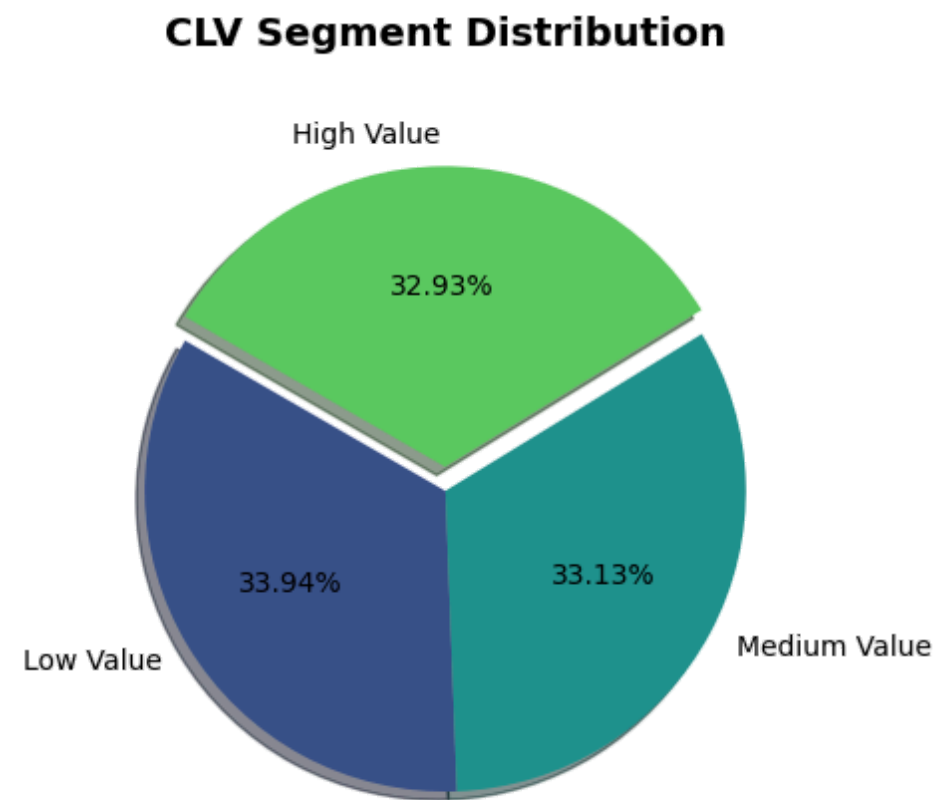


We assume COGS = 70%  
Total\_Revenue.

Top 10 High-Value Customers:

|   | AccountID | Total_Revenue | Num_Transactions | First_Purchase      | Last_Purchase       | APV        | Frequency | Lifespan_Months | COGS     | GrossMargin | CLV        |
|---|-----------|---------------|------------------|---------------------|---------------------|------------|-----------|-----------------|----------|-------------|------------|
| 0 | AC00460   | 5570.34       | 12               | 2023-01-03 17:14:14 | 2024-01-01 16:45:56 | 464.195000 | 12        | 12.066667       | 3899.238 | 0.3         | 20164.6308 |
| 1 | AC00363   | 4702.91       | 12               | 2023-01-31 17:14:32 | 2023-12-20 16:17:11 | 391.909167 | 12        | 10.733333       | 3292.037 | 0.3         | 15143.3702 |
| 2 | AC00071   | 4050.12       | 8                | 2023-01-02 17:00:33 | 2023-12-25 17:41:30 | 506.265000 | 8         | 11.900000       | 2835.084 | 0.3         | 14458.9284 |
| 3 | AC00021   | 4263.34       | 8                | 2023-01-23 16:03:04 | 2023-12-08 16:36:30 | 532.917500 | 8         | 10.633333       | 2984.338 | 0.3         | 13600.0546 |
| 4 | AC00304   | 3651.80       | 11               | 2023-01-02 16:07:17 | 2023-12-20 17:13:50 | 331.981818 | 11        | 11.733333       | 2556.260 | 0.3         | 12854.3360 |
| 5 | AC00265   | 3953.60       | 9                | 2023-01-17 16:48:10 | 2023-12-04 16:14:07 | 439.288889 | 9         | 10.666667       | 2767.520 | 0.3         | 12651.5200 |
| 6 | AC00136   | 3880.53       | 8                | 2023-01-03 18:18:14 | 2023-11-20 16:31:36 | 485.066250 | 8         | 10.666667       | 2716.371 | 0.3         | 12417.6960 |
| 7 | AC00257   | 3410.02       | 11               | 2023-01-02 16:27:18 | 2023-12-06 17:10:35 | 310.001818 | 11        | 11.266667       | 2387.014 | 0.3         | 11525.8676 |
| 8 | AC00020   | 3788.01       | 9                | 2023-01-12 17:58:56 | 2023-11-13 16:46:06 | 420.890000 | 9         | 10.133333       | 2651.607 | 0.3         | 11515.5504 |
| 9 | AC00316   | 3437.56       | 6                | 2023-01-02 16:01:14 | 2023-11-30 16:51:46 | 572.926667 | 6         | 11.066667       | 2406.292 | 0.3         | 11412.6992 |

# Appendix – CLV Analysis



## Key Insights

- Most customers have CLV **below average**, suggesting a need to **increase purchase frequency or order value**.
- The **High Value** group (~33%) should be the main focus for **retention and loyalty programs**.
- **Average CLV  $\approx$  3.8K** can serve as a **benchmark** for evaluating marketing and customer strategies.



Thank you for corporation

